



**NATIONAL
GEODETIC
SURVEY**

Defining and Accessing the Terrestrial Reference Frames of the Modernized NSRS

Presented by

Dan Gillins, PhD, PLS

Chief, Observation & Analysis Division (OAD)

Supervisory Geodesist

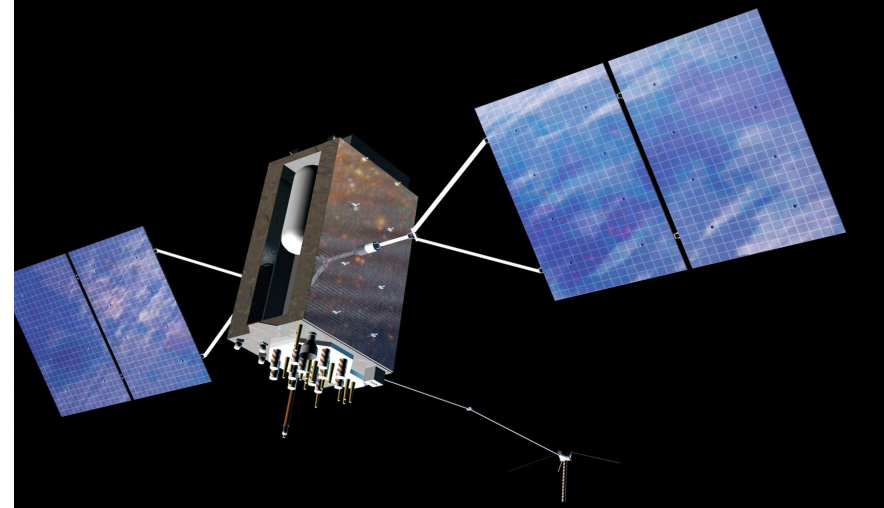


Outline

- Defining Geometric Components of the Modernized NSRS
 - ITRF2020 – why start with it?
 - Multi-Year CORS Solution 3 (MYCS3)
 - CORS Coordinate Functions
 - New plate-fixed terrestrial reference frames
 - Euler pole parameters (EPPs)
- Accessing the Terrestrial Reference Frames for the Modernized NSRS
 - BETA products
 - Modernized OPUS-Static
 - Data Delivery Service
 - NCAT

Big Changes Ahead: Modernized National Spatial Reference System (NSRS)

- Update all NSRS Coordinates
 - Replace existing datums with new datums
 - Account for coordinates changing with time
- Improve NGS products and services
 - New State Plane Coordinate System with new zones
- Make the NSRS more accurate, accessible, efficient, and compatible with GNSS
- Testing and Feedback in 2025-26



Crosswalk from Current to Modernized NSRS

CURRENT

Horizontal Datums

- NAD 83(2011)
- NAD 83(PA11)
- NAD 83(MA11)

Vertical Datums

- NAVD88
- PRVD02, VIVD08, GUV D04,
NMVD03

Hybrid Geoid Models (GEOID18)

State Plane Coordinates of 1983

Modernized

Terrestrial Reference Frames

- ITRF2020 (Global)
- NATRF2022 (North American Plate)
- PATRF2022 (Pacific Plate)
- MATRF2022 (Mariana Plate)
- CATRF2022 (Caribbean Plate)

Geopotential Datum

- NAPGD2022

Gravimetric Geoid Model (GEOID2022)

State Plane Coordinates of 2022

Understanding ITRF2020 & Its Benefits

- What is ITRF2020?

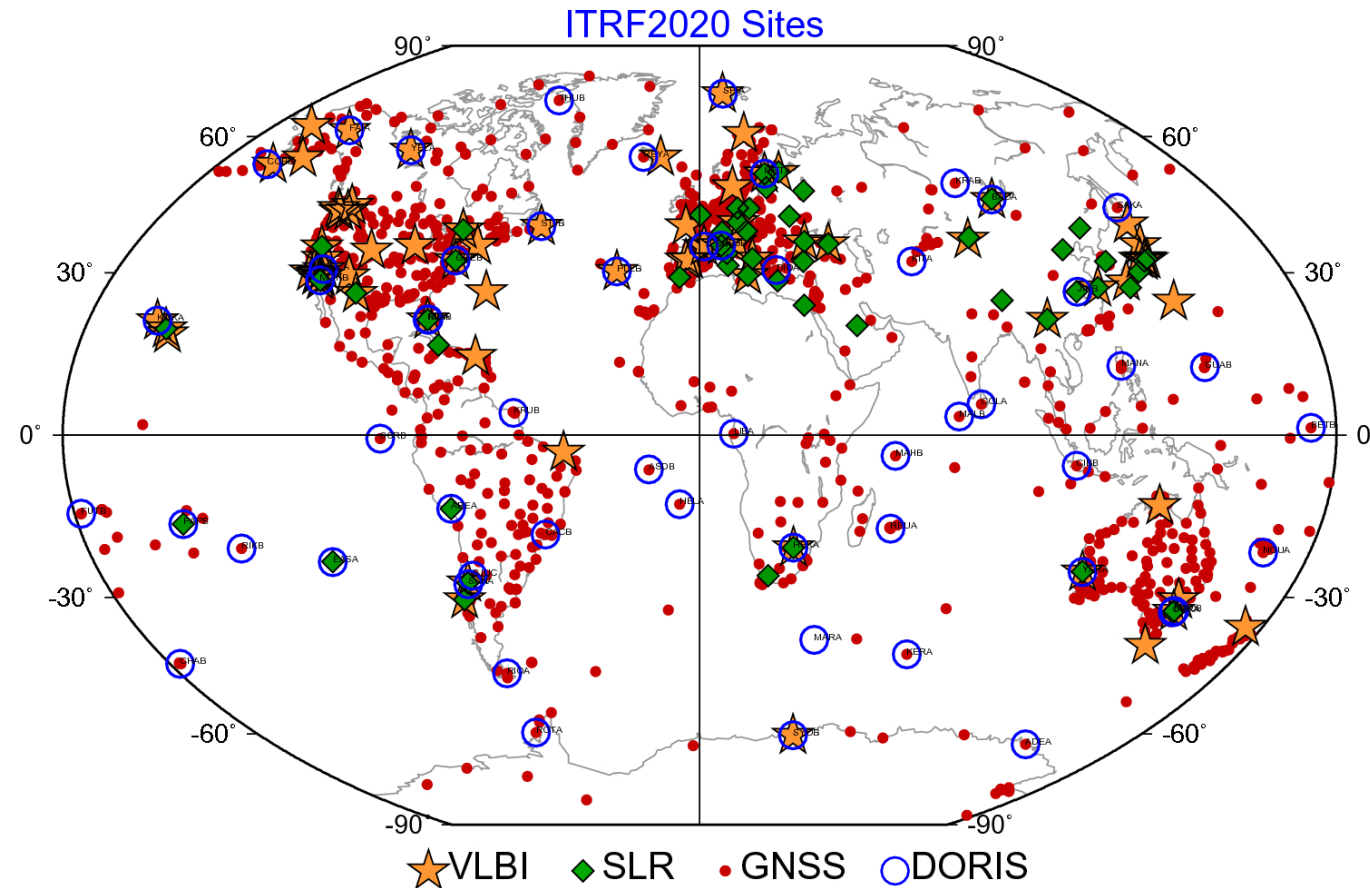
- Latest International Terrestrial Reference Frame (ITRF)
- Realized by the International Earth Rotation and Reference Systems Service (IERS)
- Precise 3D positions & velocities worldwide
- Built from GNSS, VLBI, SLR, and DORIS

- Why Does It Benefit GNSS Positioning?

- Makes coordinates globally consistent
- Improves accuracy & stability
- Supports multi-GNSS (GPS, Galileo, GLONASS, Beidou)
- Enables center-meter level precision

- Why Does it Matter?

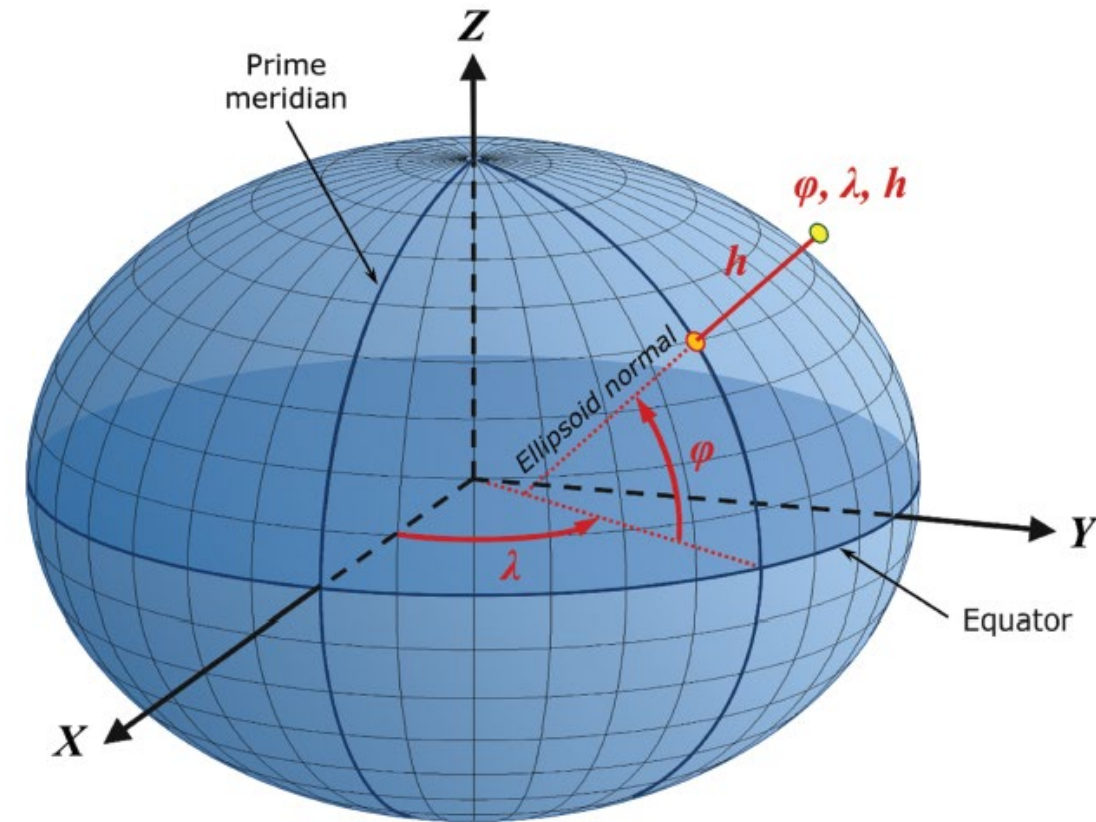
- Coordinates are time-dependent; positions change due to tectonic motion (~1-2 cm/yr in many areas)
- Foundation for the United States' upcoming Terrestrial Reference Frames
- Consistency! High-accuracy GNSS measurements and published control will all be in ITRF2020;
- Latest realization of WGS-84 is also aligned to ITRF2020



Altamimi et al. (2023)

ITRF2020: Important Concepts

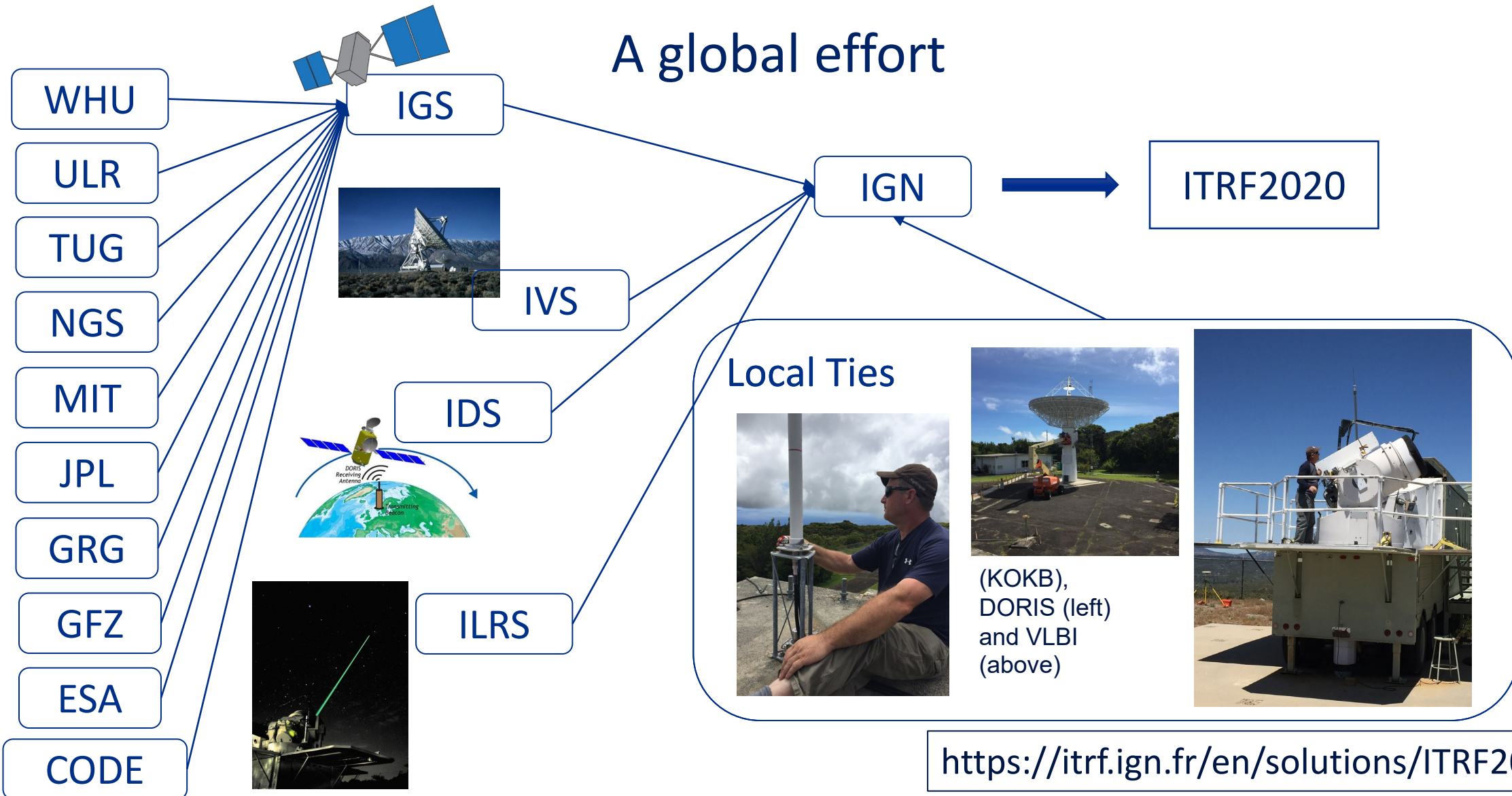
- **Origin at Earth's center of mass (geocenter)**
 - Whole Earth, including Solid Earth, Oceans, Atmosphere
 - Realized mostly by SLR
 - Accurate to the mm-level
 - GNSS satellites orbit about the geocenter
- **Orientation of the Axes**
 - Z-Axis = Earth's mean rotation axis
 - Determined using VLBI
 - X-Axis = Intersection of equator with Prime meridian
 - Y-Axis = 90 deg. east of X-axis; in equatorial plane
- **Orientation Stability**
 - Constrained so there is no net rotation relative to the Earth's crust
 - Determined by a global network of stations
 - Long-term stability ~ 0.01 milliarcseconds



Misalignment of NAD 83

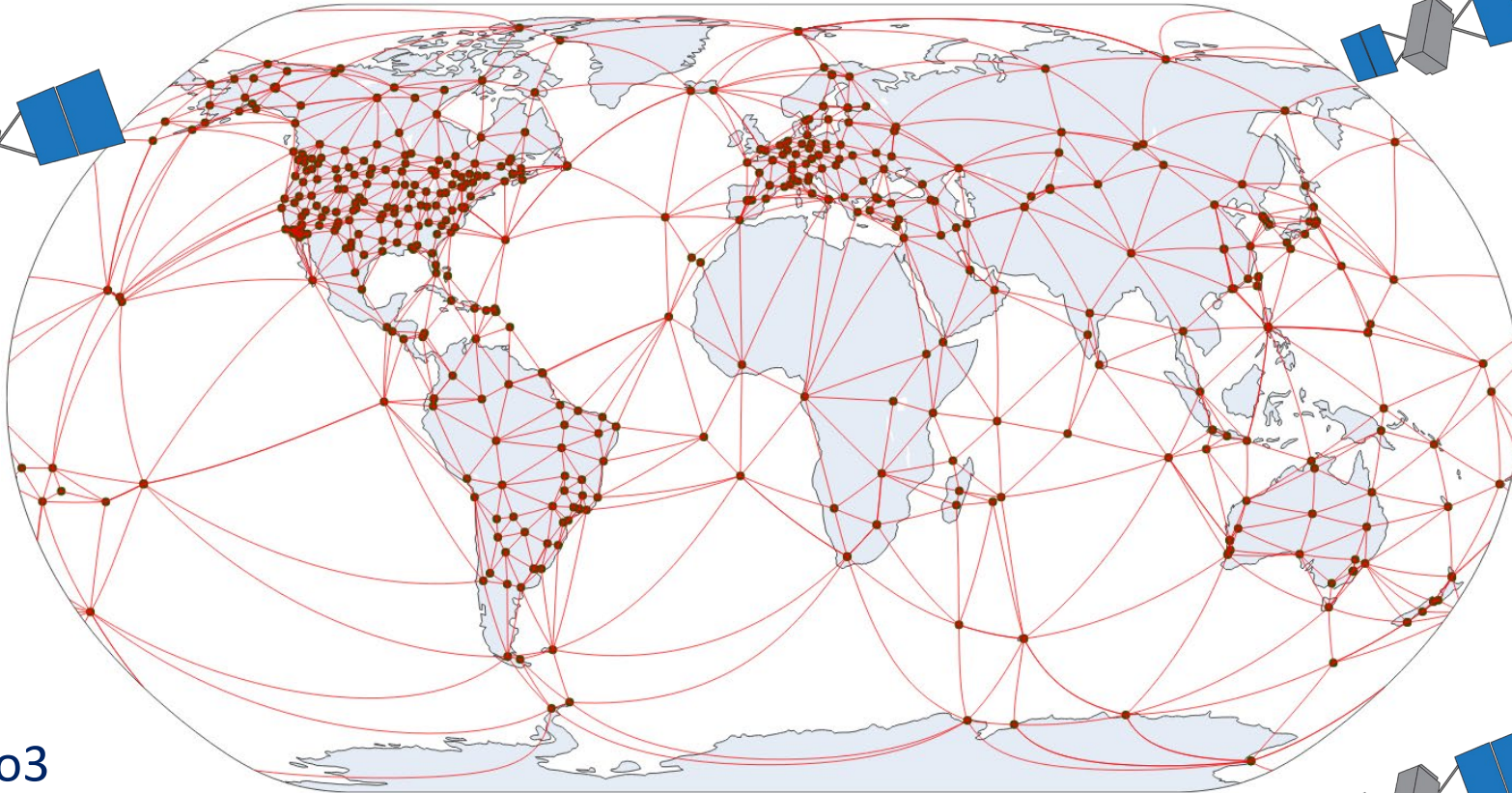
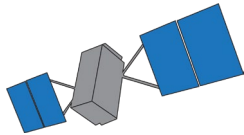
- Technically, NAD 83 is also a global reference frame
- NAD 83 was realized by terrestrial observations available in the 1970-80s
- Its origin is known to be off by ~ 2.2 m from the geocenter
- Its plate motion model for North America is rotating too fast
- NGS currently transforms GNSS measurements in ITRF2020 to NAD 83
- NGS is replacing NAD 83 with new Terrestrial Reference Frames that are built upon ITRF2020

Realization of ITRF2020



Realization of ITRF2020

NGS's contribution

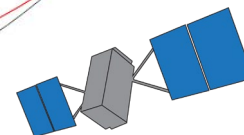


International GNSS Service
(IGS) 3rd Reprocessing
Campaign (repro3)

NGS processed data from
~800 stations

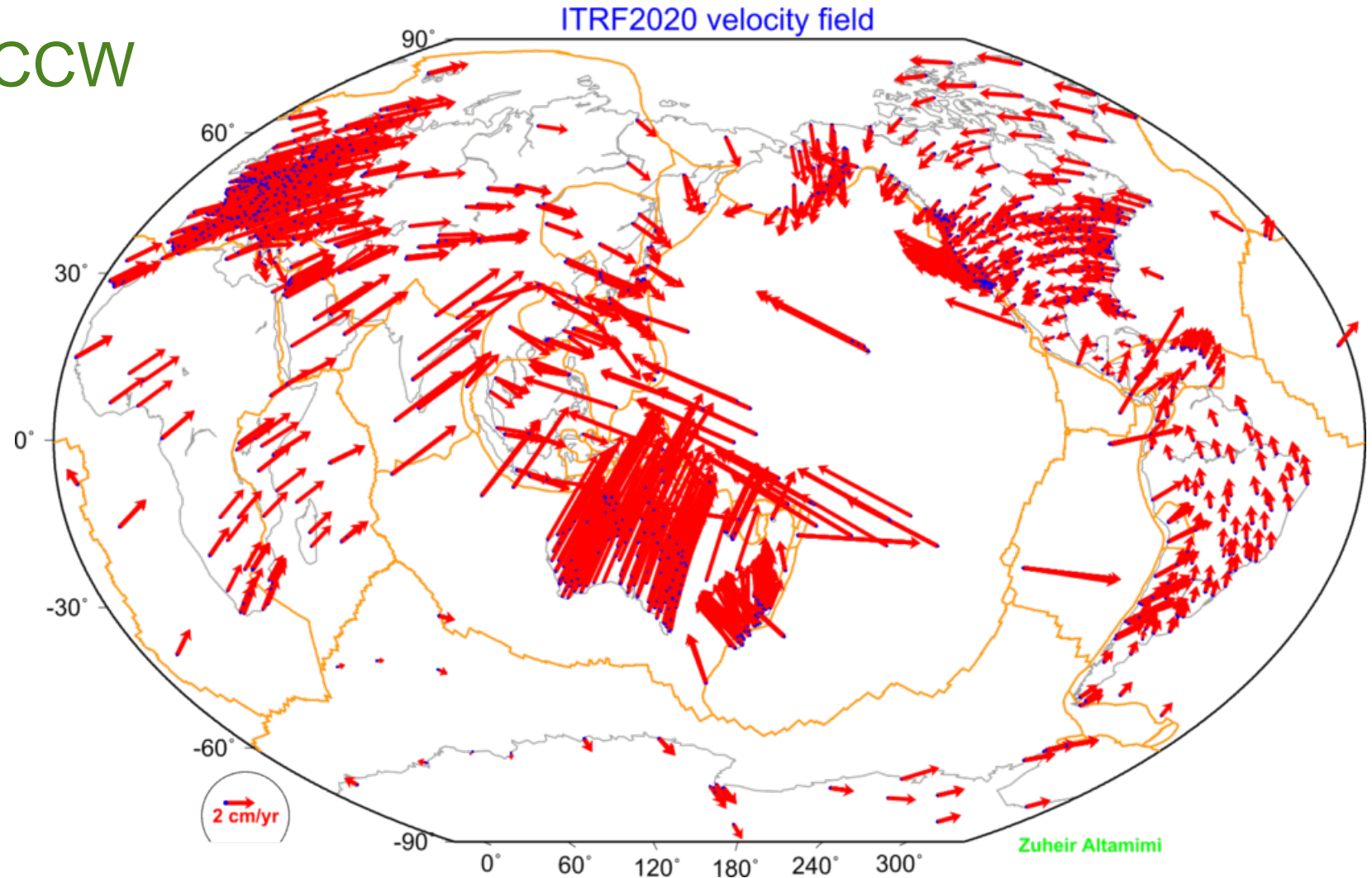
1 January 1994 through 26
November 2022 (10556 days
of data)

Repro3
2020:233



ITRF2020 Velocities at IGS Stations

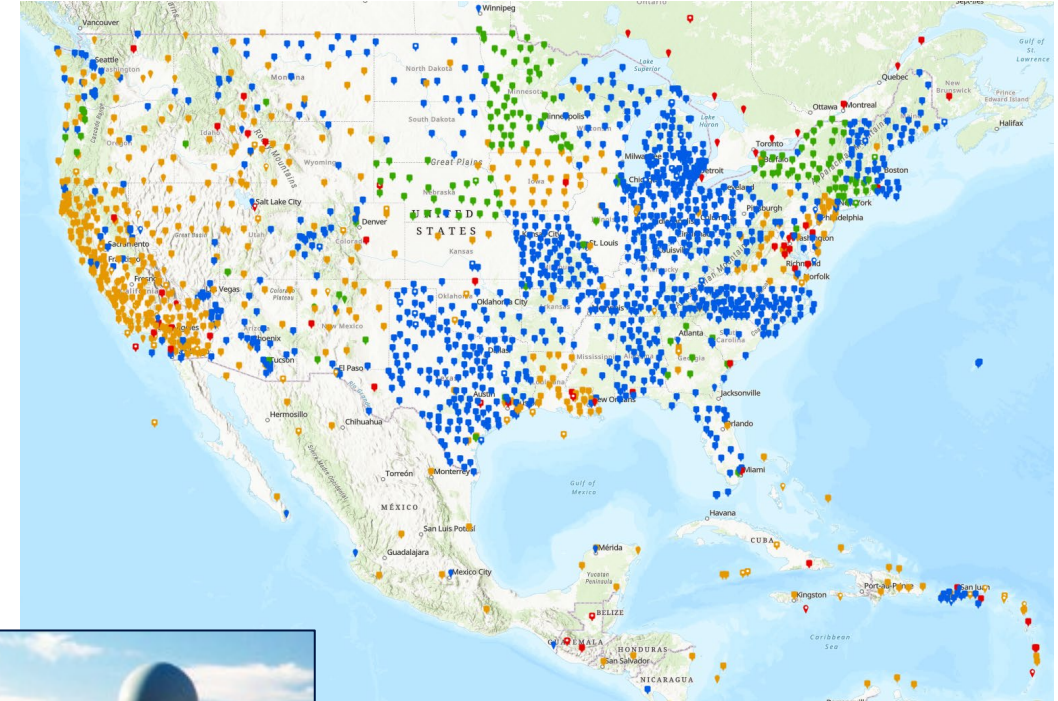
- North America Plate: CCW
- Caribbean Plate: NE
- Pacific Plate: NW
- Mariana: little info.



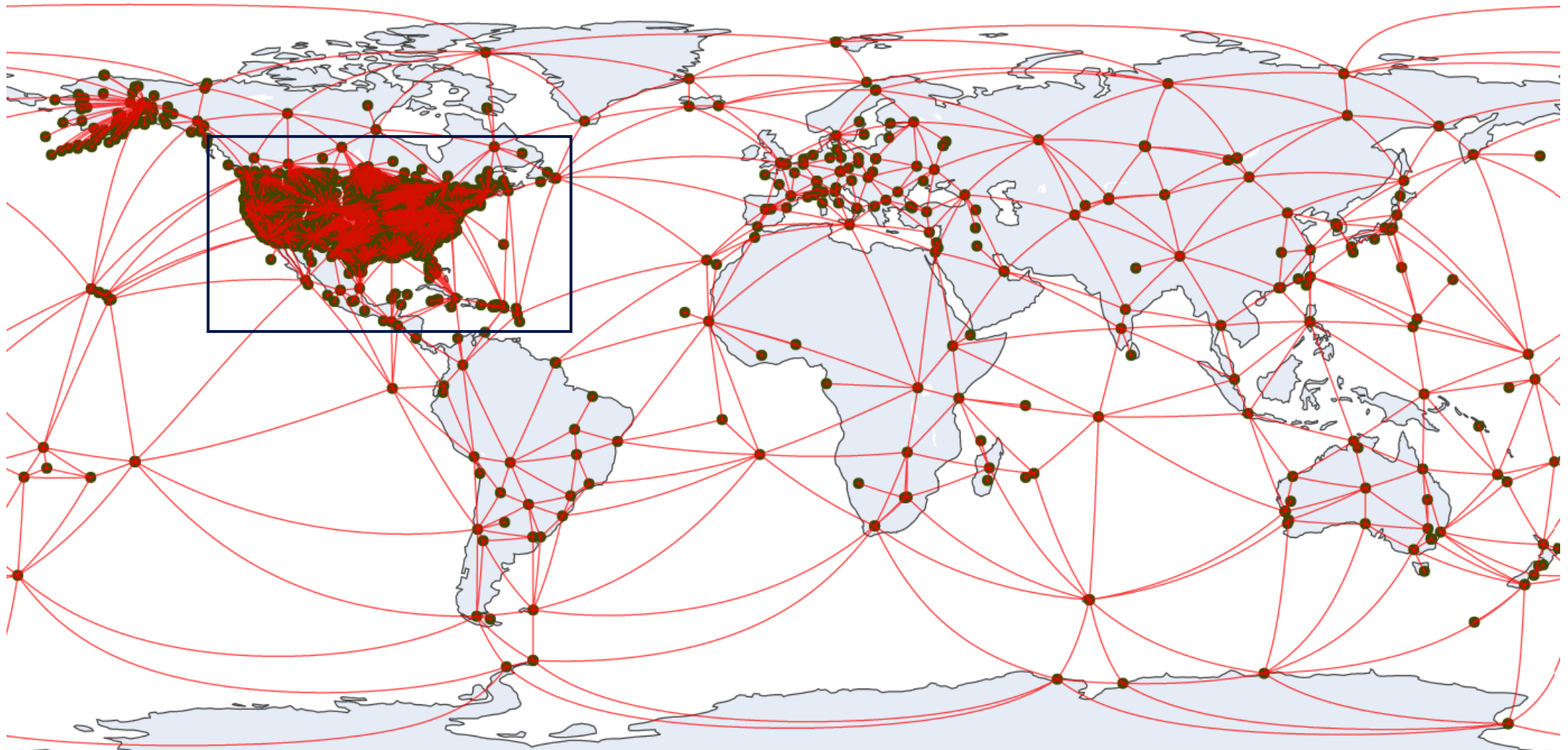
Altamimi et al. (2023)

Multi-Year CORS Solutions (MYCSs)

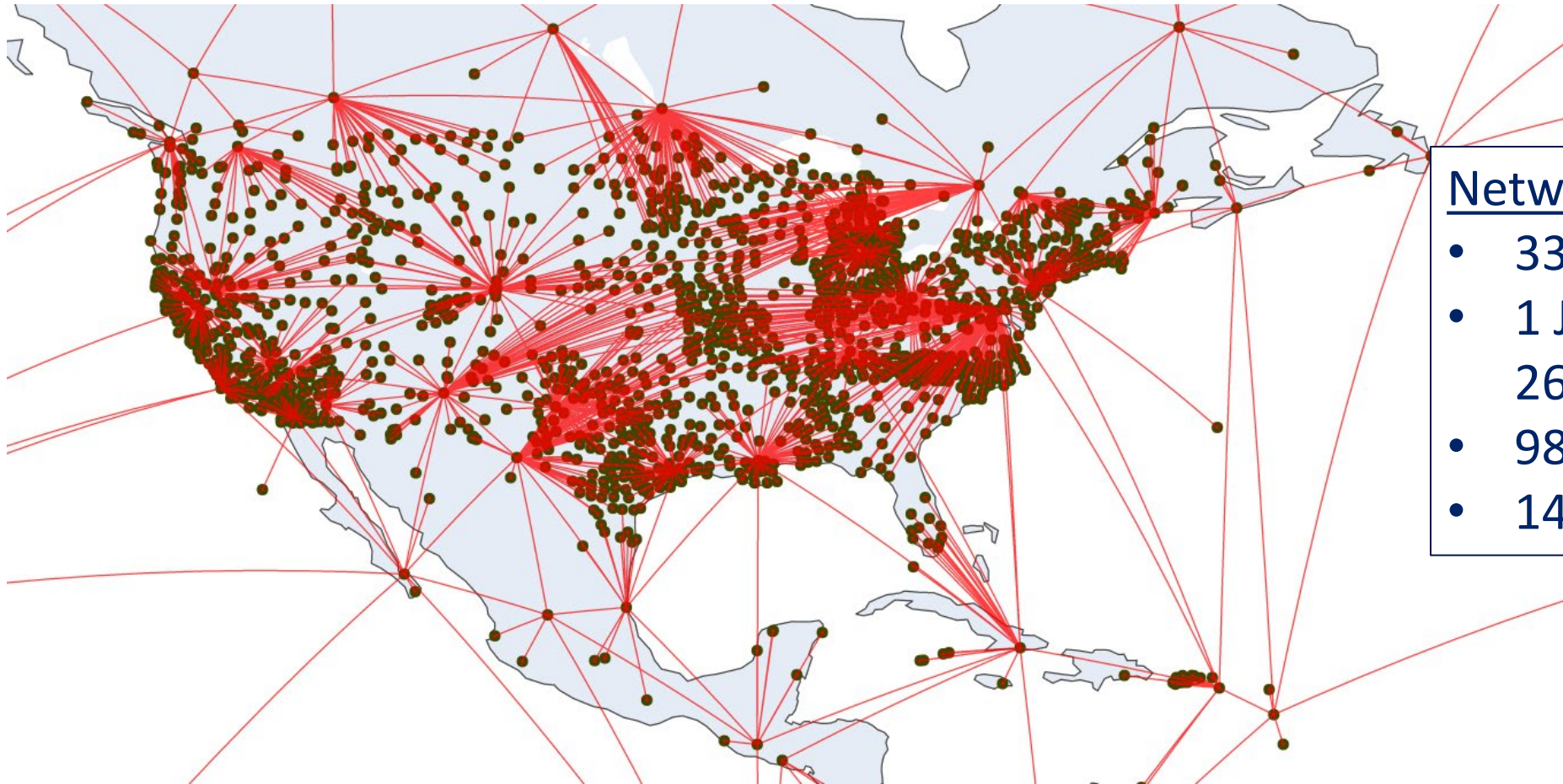
- MYCS1 = Alignment to ITRF2008
 - $t_0 = 2005.00$
 - Coordinates and velocities
- MYCS2 = Alignment to ITRF2014
 - $t_0 = 2010.00$
 - Coordinates and velocities
- MYCS3 = Alignment to ITRF2020
 - $t_0 = 2020.00$
 - Coordinates and velocities
 - Periodic (annual and semi-annual)



Aligning MYCS3 with ITRF2020



Aligning MYCS3 with ITRF2020



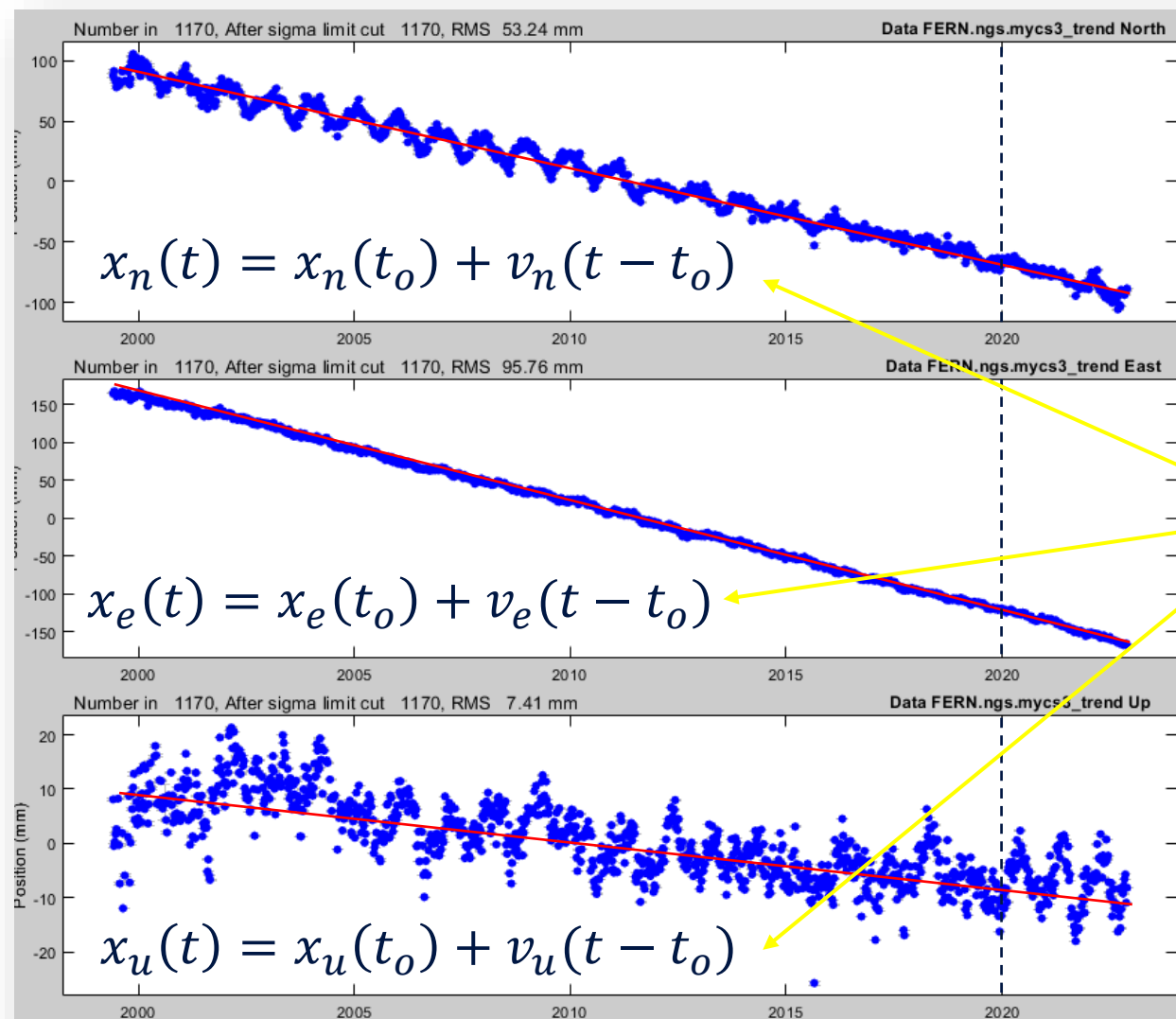
Network Processing:

- 3355 Total Stations
- 1 January 1996 through 26 November 2022
- 9827 days
- 1402 weeks

CORS Coordinate Functions



<https://www.unavco.org/instrumentation/networks/status/nota/overview/FERN>



For MYCS3
 $t_o = 2020.0$

Aligned with
ITRF2020

MYCS3 CORS Coordinate Function Model

$$\mathbf{x}(t) = \mathbf{x}_R + \mathbf{v}_x(t - t_R) + \mathbf{A}_x(2\pi t) + \mathbf{B}_x(2\pi t) + \mathbf{C}_x(4\pi t) + \mathbf{D}_x(4\pi t) + \mathbf{g}_x(t)$$

linear

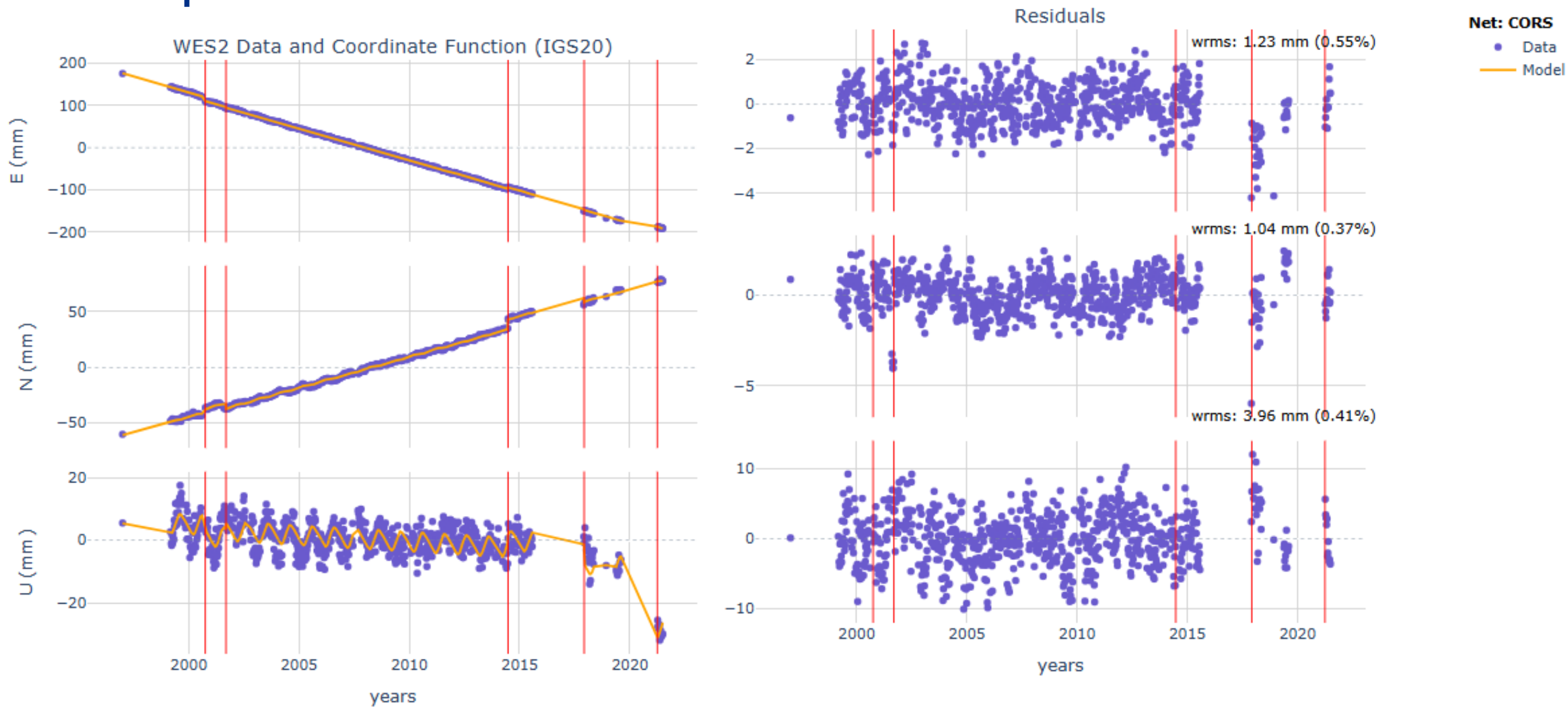
periodic (annual and semi-annual)

Parameters

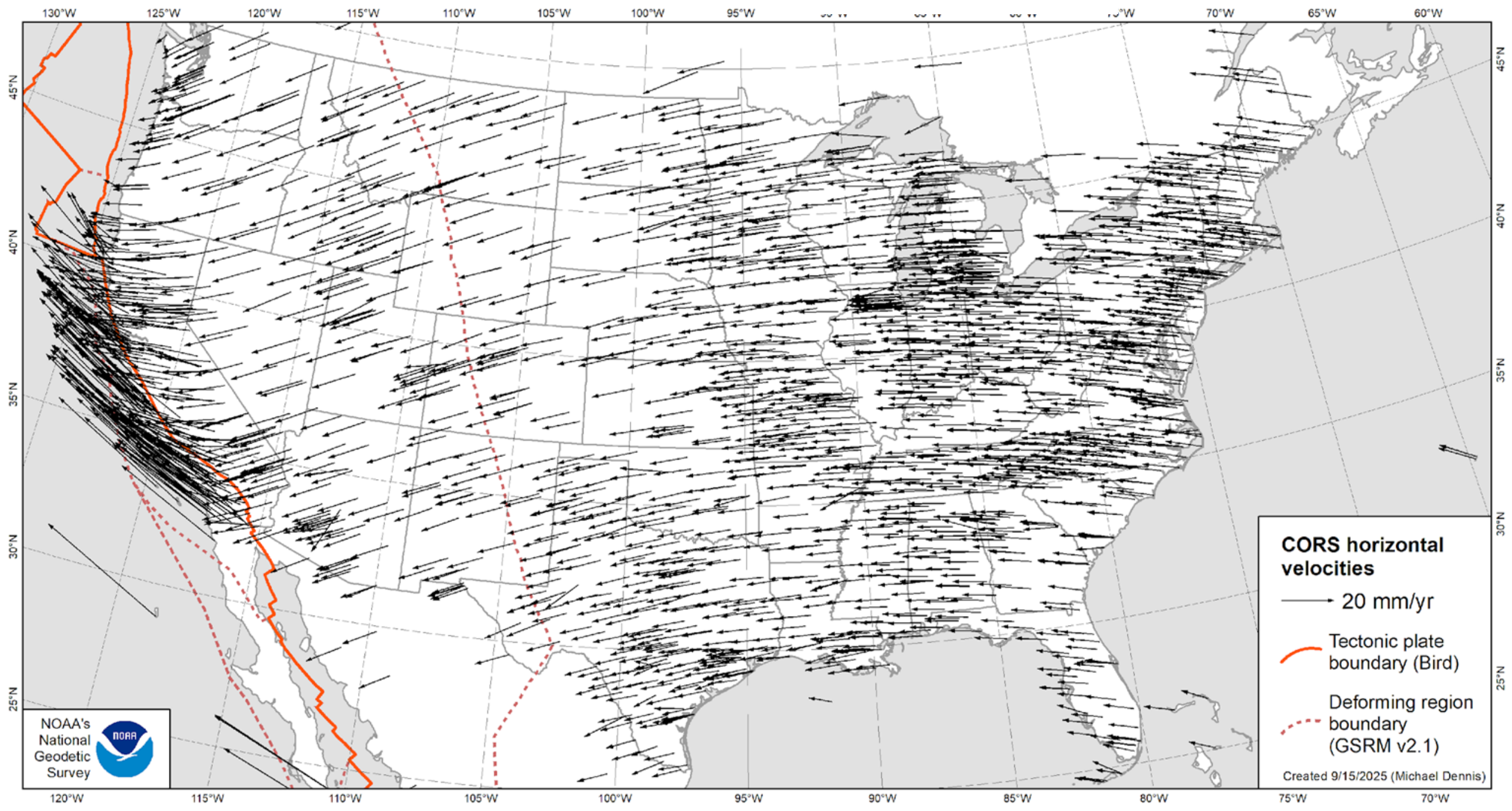
- $X, V_x, A_x, B_x, C_x, D_x = 6$ values per component times 3 components (X, Y, Z)
- 18 values per “solution”, i.e., model segment

Post-seismic deformation
Future enhancements

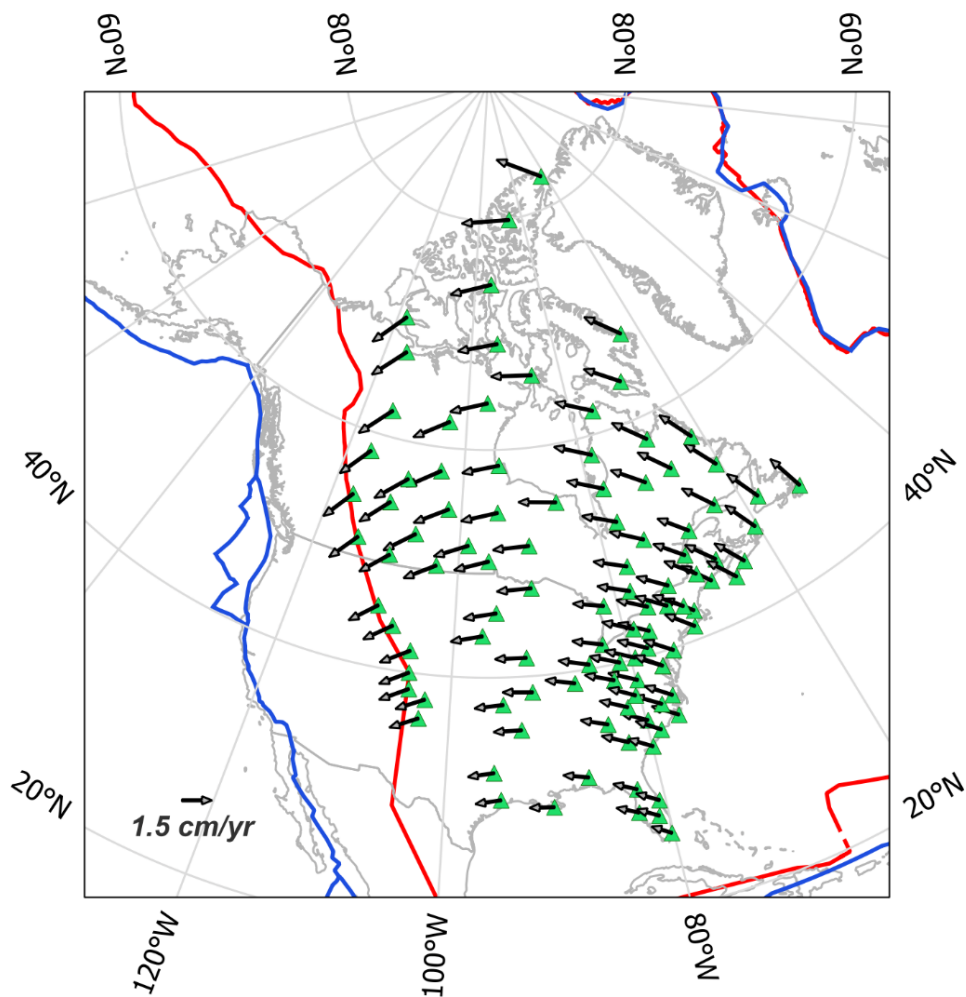
CORS Time Series – Complexities



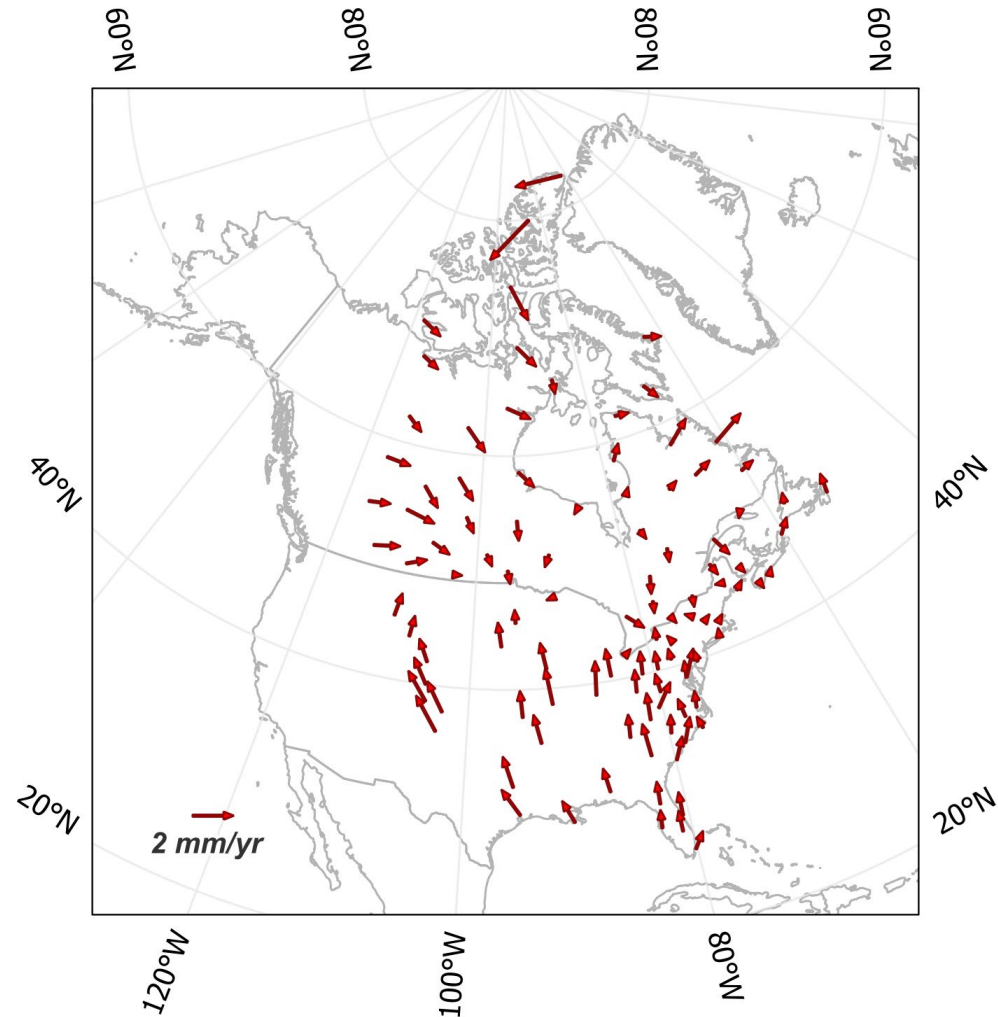
CORS horizontal velocities: In ITRF2020



Velocity Field at Frame-Defining Stations; NA



Velocities in **ITRF2020** at the frame-defining stations.



Velocities in **NATRF2022** at the frame-defining stations.

Euler Pole Parameters

$$\begin{bmatrix} X_i(t) \\ Y_i(t) \\ Z_i(t) \end{bmatrix}_{NATRF2022} = \mathbf{R}^{EPP2022}(\Delta t) \begin{bmatrix} X_i(t) \\ Y_i(t) \\ Z_i(t) \end{bmatrix}_{ITRF2020}$$

where

$$\mathbf{R}^{EPP2022}(\Delta t) \simeq \begin{bmatrix} 1 & \omega_Z \Delta t & -\omega_Y \Delta t \\ -\omega_Z \Delta t & 1 & \omega_X \Delta t \\ \omega_Y \Delta t & -\omega_X \Delta t & 1 \end{bmatrix}^{NAEPP2022}$$

$t_0 = 2020.00 \leftarrow$ reference epoch

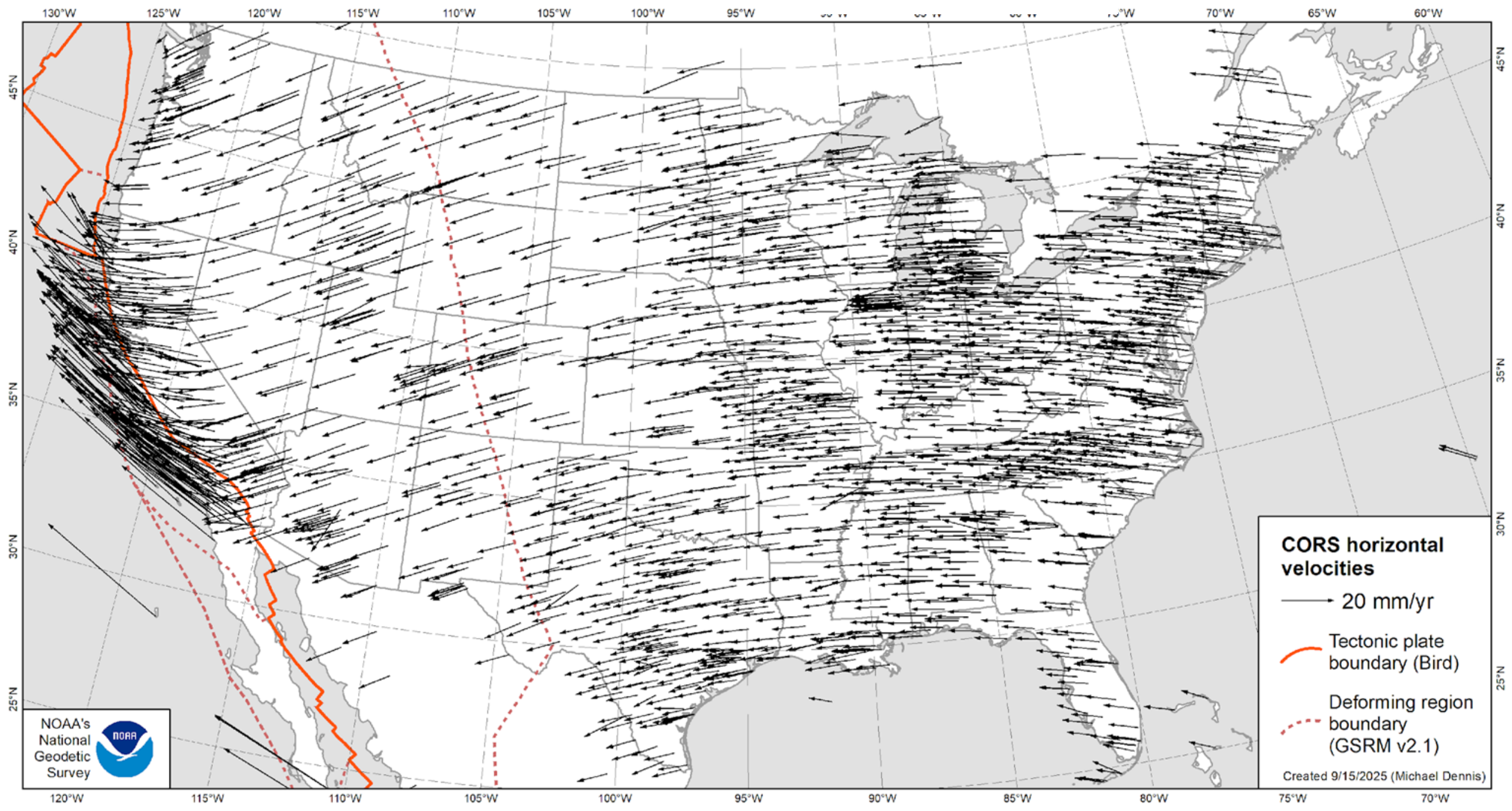
Final Euler Pole Parameters

Parameter	NATRF2022	PATRF2022	CATRF2022	MATRF2022
Latitude(°)	-3.8205	-62.511	30.803	12.1432
Longitude(°)	-86.2977	111.0382	-91.273	143.7228
Magnitude(mas/yr)	0.7075	2.467	1.202	10.264
ω_x (mas/yr)	0.046	-0.409	-0.056	-8.089
ω_y (mas/yr)	-0.704	1.063	-0.957	5.937
ω_z (mas/yr)	-0.047	-2.188	0.589	2.159

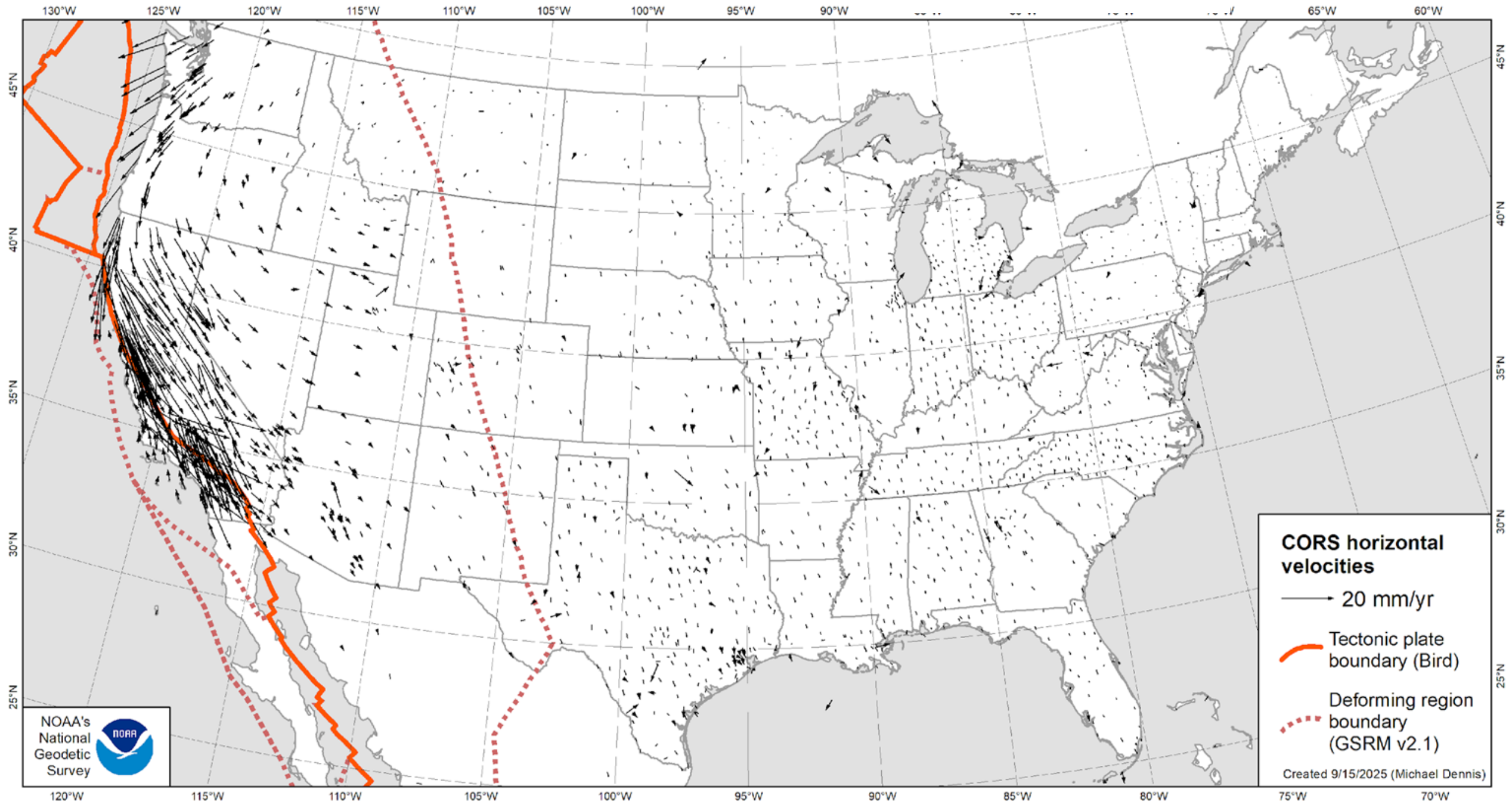
- Reference epoch(t_0): 2020.0

$$\begin{bmatrix} X_i(t) \\ Y_i(t) \\ Z_i(t) \end{bmatrix}_{\text{NATRF2022}} = \begin{bmatrix} X_i(t) + Y_i(t)\omega_z\Delta t - Z_i(t)\omega_y \Delta t \\ -X_i(t)\omega_z\Delta t + Y_i(t) + Z_i(t)\omega_x\Delta t \\ X_i(t)\omega_y\Delta t - Y_i(t)\omega_x\Delta t + Z_i(t) \end{bmatrix}^{\text{NAEPP2022}}_{\text{ITRF2020}}$$

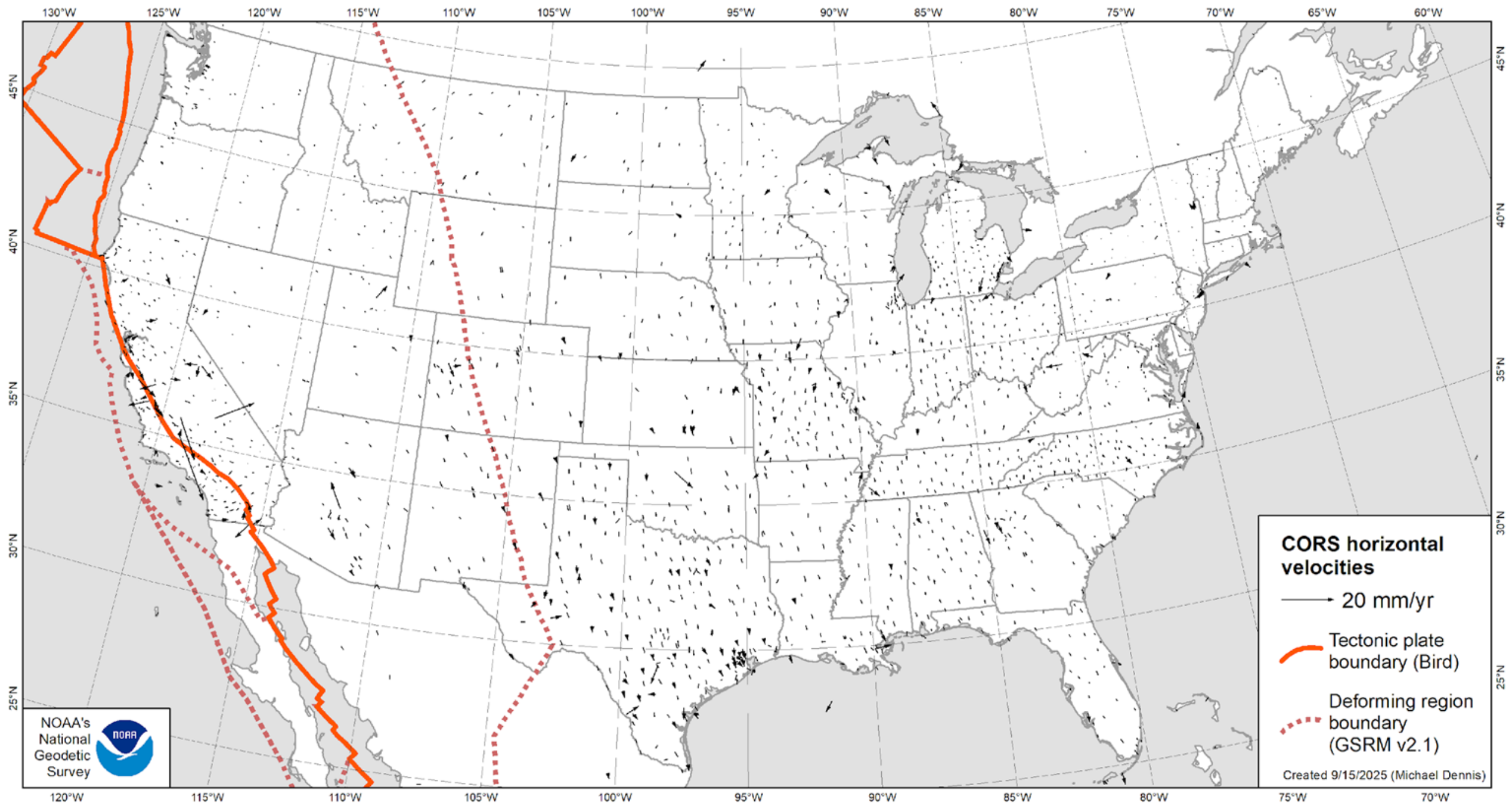
CORS horizontal velocities: In ITRF2020



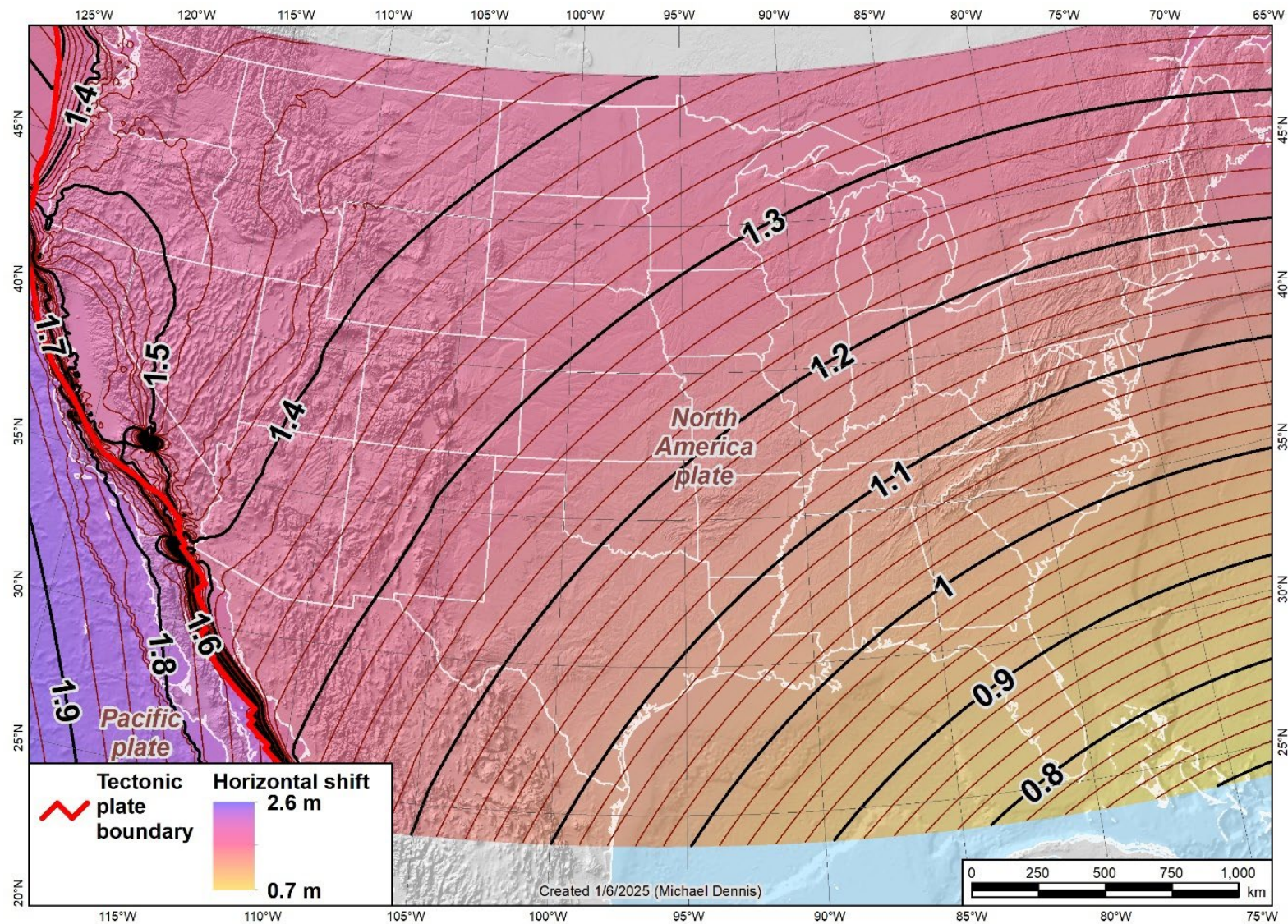
NATRF2022 Velocities; NOAA CORS Network



Residual Velocities After Applying IFDM2022; NOAA CORS Network



Horizontal Shift from NAD 83 to NATRF2022



Horizontal shift between NAD83 (2011)2010.00 and NATRF2022 (epoch 2020.0)

Summary of Steps for Positioning in the Modernized NSRS

1. Process GNSS baselines at the time of the measurements (e.g., 2026.25) using latest ITRF2020 products
2. Reduce GNSS data into dX, dY, dZ vector components
3. Transform results to ITRF2020 at epoch 2020.00 by applying IFDM2022 (deformation grid)
4. Constrain published control (e.g., CORS) coordinates in ITRF2020 epoch 2020.00
5. Perform least squares adjustment. Report adjusted coordinates

Note: since $\Delta t = 0$, coordinates in ITRF2020 = NATRF2022 at epoch 2020.00!

Outline

- *Defining Geometric Components of the Modernized NSRS*
 - *ITRF2020 – why start with it?*
 - *Multi-Year CORS Solution 3 (MYCS3)*
 - *CORS Coordinate Functions*
 - *New plate-fixed terrestrial reference frames*
 - *Euler pole parameters (EPPs)*
- **Accessing the Terrestrial Reference Frames for the Modernized NSRS**
 - BETA products
 - Modernized OPUS-Static
 - Data Delivery Service

Beta Products

<https://beta.ngs.noaa.gov/>

- **NAPGD2022**
 - GEOID2022
 - Surface gravity model
 - Deflection of vertical model
- **EPPs for the 4 plate-fixed terrestrial reference frames**
 - NATRF2022, PATRF2022, MATRF2022, CATRF2022
- **State Plane Coordinate System of 2022**
 - Zone definitions and zone polygons
- **NCAT – early version for SPCS2022**

BETA Products

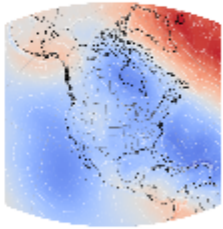
Welcome to the NGS National Spatial Reference System (NSRS) Beta Product Release Site. The products and tools on this site represent ready-to-be tested components of the modernized NSRS. The purpose of this site is to hold products and tools supporting the modernized NSRS through their roll-out and testing (2024-2026).

Products on this site are preliminary releases that are for testing and evaluation only; they are not final products nor do they contain any authoritative NGS data or tools. To provide feedback on any of the content on this site, please email ngs.feedback@noaa.gov.

NAPGD2022

The North American-Pacific Geopotential Datum of 2022 will serve as the future geopotential datum within the NSRS. It covers three regions: the North American-Pacific region, Guam and Commonwealth of Northern Mariana Islands, and American Samoa. This datum contains all of the necessary information to provide mutually consistent orthometric heights, geoid undulations, gravity anomalies, deflections of the vertical and other geodetic coordinates related to the gravity field.

[Preview](#)



NATRF2022, PATRF2022, CATRF2022, & MATRF2022

Four new terrestrial reference frames will serve as the future reference for geodetic latitude, longitude, and ellipsoidal height within the NSRS. They are defined by transformation from the International Terrestrial Reference Frame of 2020 (ITRF2020), the frame used for high-accuracy GNSS positioning, using Euler Pole Parameters that account for tectonic plate motion for the North America (NATRF2022), Pacific (PATRF2022), Caribbean (CATRF2022), and Mariana (MATRF2022) tectonic plates.

[Preview](#)



State Plane Coordinate System of 2022 (SPCS2022)

Preliminary release of the third generation of the State Plane Coordinate System. SPCS2022 will be referenced to the 2022 NSRS Terrestrial Reference Frames to support surveying, engineering, and mapping activities in the Modernized NSRS.

[Preview](#)



NGS Coordinate Conversion and Transformation Tool (NCAT)

A preliminary version of NCAT that supports coordinate conversions for the preliminary State Plane Coordinate System of 2022. Transformations between existing and modernized NSRS reference frames and datums will be added to a later version.

[Preview](#)



EPPs for Plate-Fixed Terrestrial Reference Frames—Now Final!

BETA PRODUCTS ▾

[Home](#) > NATRF2022, PATRF2022, CATRF2022, and MATRF2022

NATRF2022, PATRF2022, CATRF2022, and MATRF2022

Four new terrestrial reference frames will replace the North American Datum of 1983 (NAD 83). They will serve as a foundational component of the modernized NSRS, providing precise definitions for geodetic latitude, longitude, and ellipsoidal height.

All of the frames are identical to the International Terrestrial Reference Frame of 2020 (ITRF2020) at epoch 2020.0 (January 1, 2020). As time elapses from 2020.0, each reference frame rotates with the stable part of their tectonic plate, thereby minimizing changes in coordinates due to plate motion.

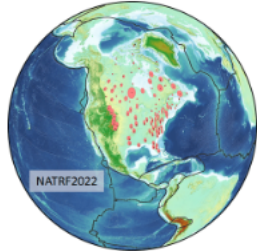
Transformations from ITRF2020 to the reference frames for the North America (NATRF2022), Pacific (PATRF2022), Caribbean (CATRF2022), and Mariana (MATRF2022) tectonic plates are made using Euler Pole Parameters (EPPs). The Beta EPPs in this table were derived using velocities relative to ITRF2020.

PLATE	Omega X (mas/yr)	Omega Y (mas/yr)	Omega Z (mas/yr)
NATRF2022	0.046	-0.704	-0.047
CATRF2022	-0.056	-0.957	0.589
PATRF2022	-0.409	1.063	-2.188
MATRF2022	-8.089	5.937	2.159



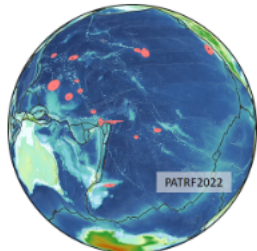
NATRF2022

NATRF2022 Euler pole parameters transform coordinates from the ITRF2020 frame of the NSRS to the North America plate frame. Coordinates of points located within the stable interior of this tectonic plate should evolve slowly with respect to the NATRF2022 frame. However, many points will exhibit residual motions associated with plate boundary zone deformation in the west, glacial isostatic adjustment, localized subsidence, or other causes.



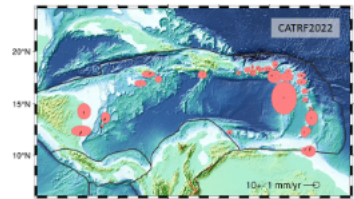
PATRF2022

PATRF2022 Euler pole parameters transform coordinates from the ITRF2020 frame of the NSRS to the Pacific plate frame. Coordinates of points located within the stable interior of this tectonic plate should evolve very slowly with respect to the PATRF2022 frame. Unlike the North American and Caribbean plates, the Pacific plate does not experience appreciable large-scale strain. Points very near plate boundaries or on volcanic islands, such as the big island of Hawaii, may exhibit relatively localized residual motions.



CATRF2022

CATRF2022 Euler parameters transform coordinates from the ITRF2020 frame of the NSRS to the Caribbean plate frame. Coordinates of points located within the stable interior of this Caribbean plate should evolve slowly with respect to the CATRF2022 frame. However, many points in the Caribbean region are located near active faults along the periphery of the plate, and others are located on volcanically active islands. Therefore, some stations are expected to exhibit residual motions associated with plate boundary zone or volcano deformation.



MATRF2022

MATRF2022 transforms coordinates from the ITRF2020 frame of the NSRS to the Mariana plate frame. Coordinates of points located within the stable interior of this tectonic plate should evolve slowly with respect to the MATRF2022 frame. However, many points in the volcanically active Mariana Islands region are located along the margin of the Mariana. Therefore, some points may exhibit residual motions associated with plate boundary zone or volcano deformation. The Mariana plate is a small, crescent shaped plate between the Pacific and Philippine plates. Guam and the Commonwealth of the Northern Mariana Islands (CNMI) are on this plate and home to many US citizens and interests. Given its small size, no good visual is available as for the other three plates.

Upcoming Beta Releases

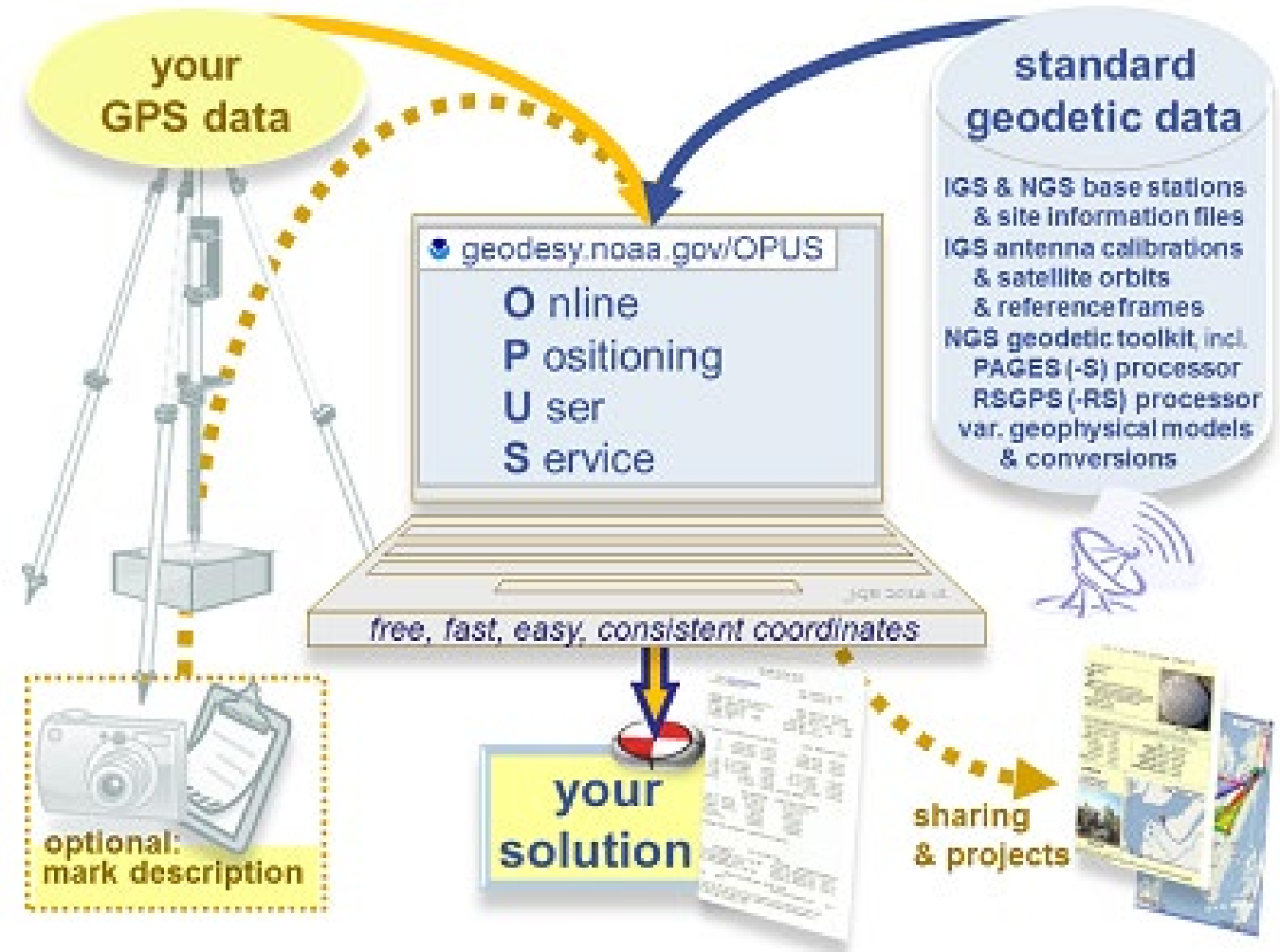
- **OPUS-S 5.2**
 - OPUS-Static solutions in Modernized NSRS
 - Multi-GNSS capable
 - Sharing possible
- **Data Delivery Service**
 - Replacement to Datasheets
 - Geodetic Mark Page (e.g., passive marks)
 - Geodetic Station Page (CORS)
- **NCAT**
 - Coordinate transformations
 - NADCON and VERTCON grids
 - Deformation modeling (IFDM2022)

Tools For
Access!!

Brief Background on OPUS

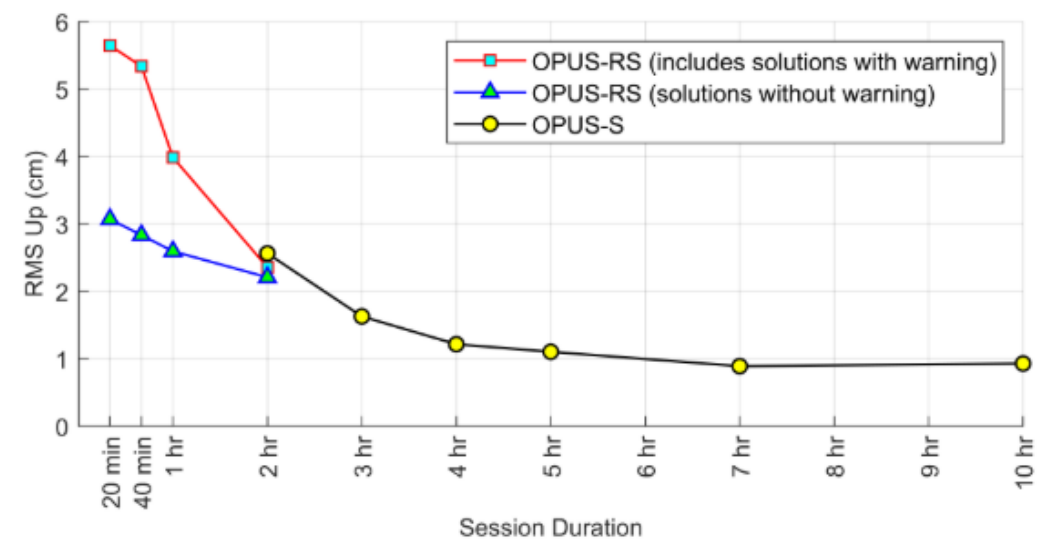
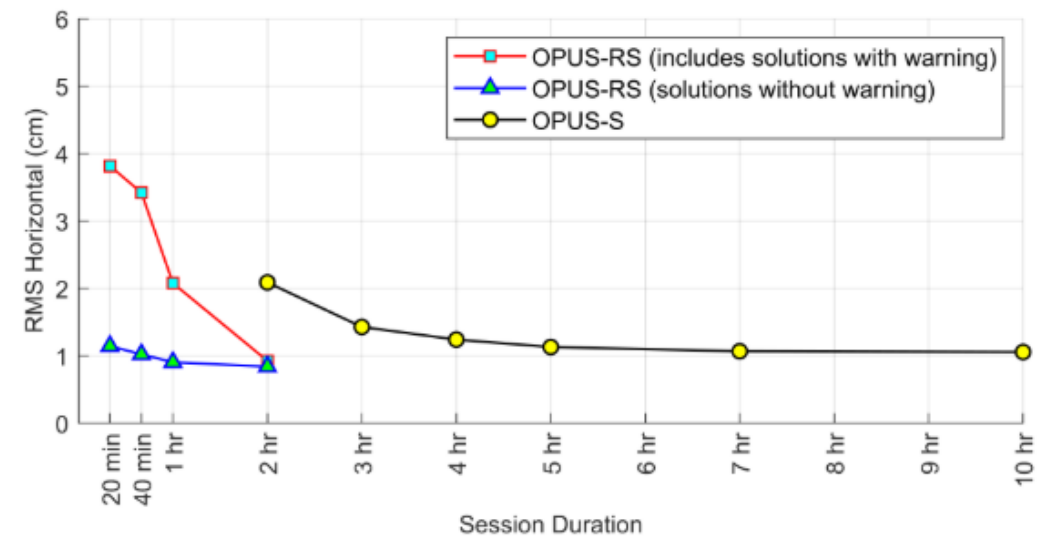
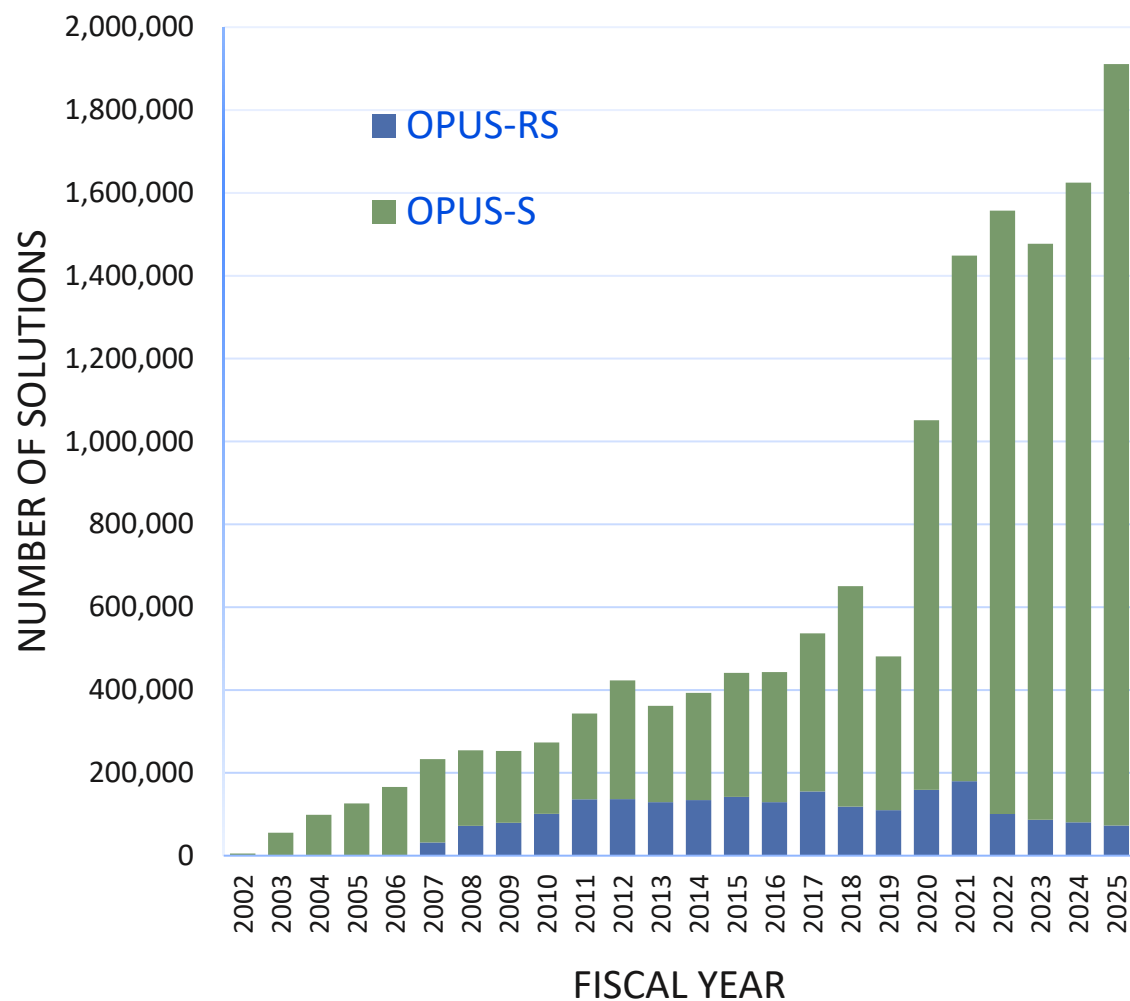
<https://geodesy.noaa.gov/OPUS/>

- Online tool, freely available
- Upload dual-frequency (L1/L2) GNSS data
- Emails coordinates or “solution” in NSRS
- **OPUS-Static (OPUS-S)**
 - References to 3 base stations
 - Sequential baseline processing (PAGES)
 - Averages 3 baselines
 - Minimum $T \geq 2$ hours
 - If $T \geq 4$ hours and meets quality thresholds, option to “Share” solution
- **OPUS-Rapid Static (OPUS-RS)**
 - References 3 – 9 base stations
 - Session baselines processing (RSGPS)
 - Minimum $T \geq 15$ min, Maximum $T < 2$ hours
- **OPUS-Projects 5**
 - Campaign-style survey projects
 - Multiple occupations on a network of marks
 - Options for designing sessions
 - RTK vectors
 - Network adjustments
 - Submission to NGS for publication



Utility and Accuracy of OPUS-S, OPUS-RS

OPUS SOLUTIONS PER YEAR



Upcoming OPUS Product Changes

July 2025: NGS Announced that in the Modernized NSRS

- OPUS-S
 - Functions similar to today's tool
 - Multi-GNSS
 - Sharing enabled
 - Capable of processing shorter occupations
- OPUS-RS
 - Will not be modernized
 - Discontinued
- OPUS-Projects 5
 - Will not be modernized
 - Discontinued
 - 6 months' notice will be given for users to submit their surveys for publication
- OPUS 6
 - Launch a new development to replace OPUS Projects 5 functionality

NGS News

Rollout Plan for Remaining NSRS Modernization products, including OPUS Products Changes

On June 17, 2025, NGS released [the first preliminary products of the modernized National Spatial Reference System \(NSRS\)](#) for beta testing and feedback. In the coming months, additional products listed below will be made available. As each product is released, it will undergo at least six months of testing preceding the final adoption and implementation of the modernized NSRS.

The descriptions below supersede previous updates or information shared in NSRS Modernization blueprint documents, plans, or presentations. These products and their status will be described on the [Track Our Progress webpage](#).

1. The **Data Delivery System (DDS)** landing page will provide an updated version of the "NGS Map" and "Looking for Benchmarks" pages. This new landing page will allow you to access modernized informational pages about geodetic stations and geodetic marks.
2. **Geodetic station pages** will offer an updated version of the current NOAA CORS Network (NCN) station pages. **Geodetic mark pages** will be updated datasheets,

Development Phases for OPUS

OPUS-S 5.2

- **Purpose:** Update OPUS-S to give access to Modernized NSRS
- **New Features:**
 - Multi-GNSS
 - Solutions in Modernized NSRS
 - OPUS-Share supported
 - $T \geq 1$ hour
- **Status:** Development Completed, Internal testing

OPUS 6.0

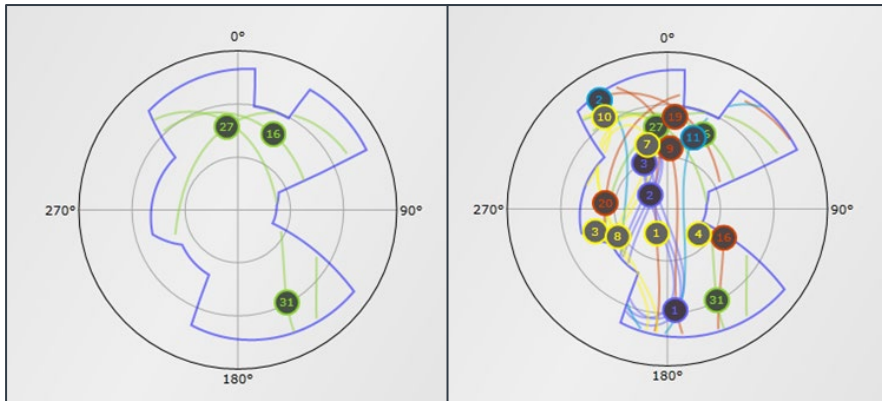
- **Purpose:** Re-write of OPUS in modern programming language
- **New Features:**
 - Cloud-native application
 - Single/multiple occupations on a mark
 - Preprocessing
 - Enhanced sharing functionality
- **Status:** Project plan & requirements completed; development underway

OPUS 6.1

- **Purpose:** Replacement for OPUS-Projects functionality
- **New Features:**
 - Multiple occupations on network of marks
 - Session processing customization
 - Network adjustments
- **Status:** Requirement gathering

M-PAGES

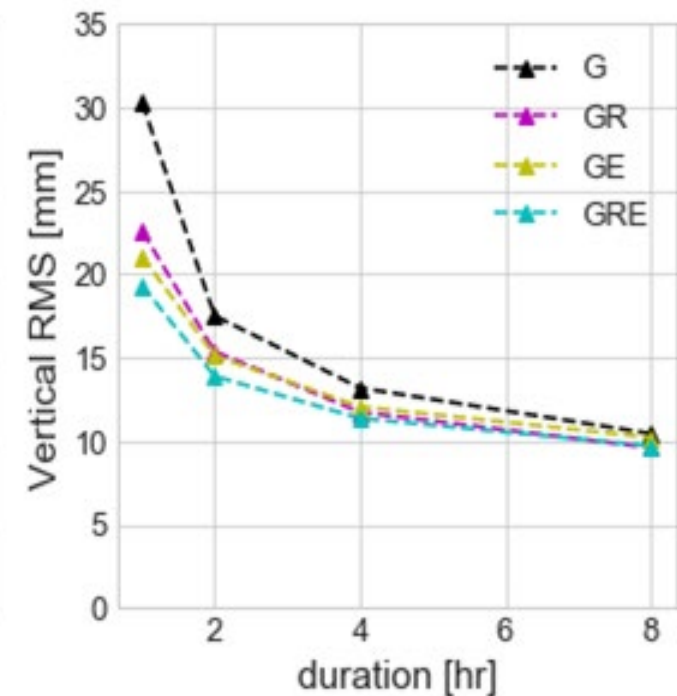
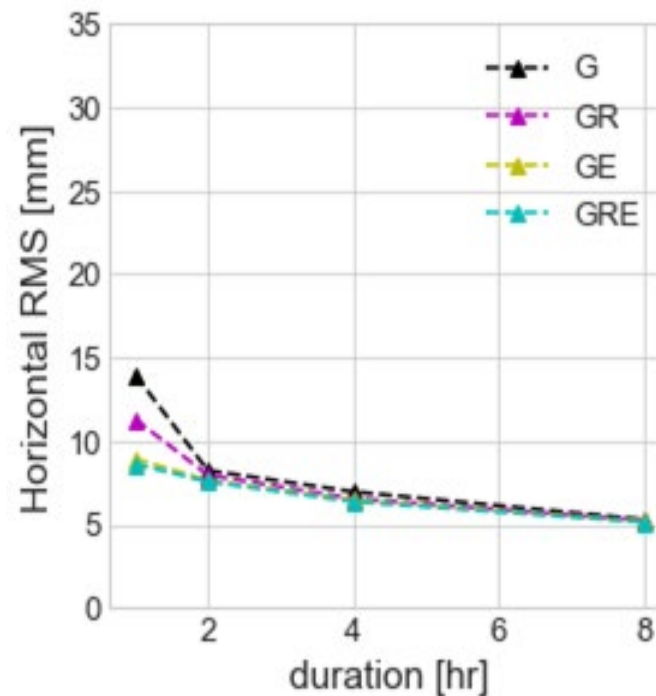
- Replacement to GPS-only software, PAGES
- GNSS data for all dual-frequency systems
- More satellites = higher accuracy
- Single differencing



<http://www.gnssplanningonline.com>



G, GPS
R, GLONASS
E, Galileo



From Stressler (2022)

OPUS-S 5.2: Multi-GNSS Capabilities

Upload your data file.

Solve your GNSS position & tie it to the National Spatial Reference System.

What is OPUS? FAQs

No file chosen

data file of dual-frequency GNSS observations. [sample](#)

antenna - choosing wrong may degrade your accuracy.

meters above your mark.

antenna height of your antenna's reference point.

email address - your solution will be sent here. [Privacy Act Statement](#)



constellations
allowed

☒GPS ☒GLONASS ☒GALILEO ☐BEIDOU ☐QZSS

formats

base stations

identify any CORS you wish to explicitly 'Use' or 'Exclude' from your solution using the buttons

find CORS IDs

Use:

1LSU00USA
1NSU00USA
1ULM00USA
AB0100USA
AB0200USA
AB0400USA

Exclude:

state plane
explained

share my
solution

OPUS-S 5.2: Example Standard Report

GENERAL INFO

RINEX File : 2ref277q.23o
RINEX Version : 2.11
User : XXXXXXXXXX
Date of Submission : 2026-02-10 13:44:32 UTC
Start of Observation : 2023-10-04 16:59:00 GPS (DOY 277)
End of Observation : 2023-10-04 23:59:30 GPS (DOY 277)
Duration of Observation : 07:00:30 (hh:mm:ss)

PROCESSING INFO

Software : M-PAGES 1.1.1 OPUS-S 5.2.0 250701
Ephemeris : igs22823.eph
Processing Mode : Dual-frequency single-difference iono-free
Data Interval : 30.0 (s)
Elevation Angle Cutoff : 10.0 (deg)
GNSS Requested : GRE

G, GPS
R, GLONASS
E, Galileo

EQUIPMENT INFO

Receiver Name : Trimble NetR5
Antenna Name : TRM41249.00 NONE
ARP Height : 1.9997 (m)
Antenna Calibration File : ngs20.atx

BASELINE BASE STATIONS AND STATISTICS

PID	STATION ID	LENGTH (km)	RMS (m)	GNSS	FIXED AMB	OBS USED
DM5409	NCBX00USA	356.4326	0.0149	GR	77.55	22980.00
AJ2122	CORB00USA	112.9484	0.0131	GR	80.85	23459.00
DJ8885	BACO00USA	113.9848	0.0150	GR	78.00	23166.00

OPUS-S 5.2: Example Standard Report (cont.)

ESTIMATED COORDINATES: ITRF2020 (EPOCH: 2023.7584)

 X: 1199922.734 (m)
 Y: -4855715.522 (m)
 Z: 3944305.003 (m)
 Std. Dev. X: 0.0035 (m)
 Std. Dev. Y: 0.0038 (m)
 Std. Dev. Z: 0.0027 (m)

Lat: N 038 26 45.20847 (ddd mm ss.sssss)
 Lon: W 076 07 9.86840 (ddd mm ss.sssss)
 Eht: -35.375 (m)
 Std. Dev. Lat: 0.0030 (m)
 Std. Dev. Lon: 0.0035 (m)
 Std. Dev. Eht: 0.0036 (m)

ESTIMATED COORDINATES: NATRF2022 (EPOCH: 2020.0000)

 X: 1199922.790 (m)
 Y: -4855715.521 (m)
 Z: 3944304.994 (m)
 Std. Dev. X: 0.0035 (m)
 Std. Dev. Y: 0.0038 (m)
 Std. Dev. Z: 0.0027 (m)

Lat: N 038 26 45.20800 (ddd mm ss.sssss)
 Lon: W 076 07 9.86616 (ddd mm ss.sssss)
 Eht: -35.371 (m)
 Std. Dev. Lat: 0.0030 (m)
 Std. Dev. Lon: 0.0035 (m)
 Std. Dev. Eht: 0.0036 (m)

ESTIMATED HEIGHT: NAPGD2022 (EPOCH: 2020.0000)

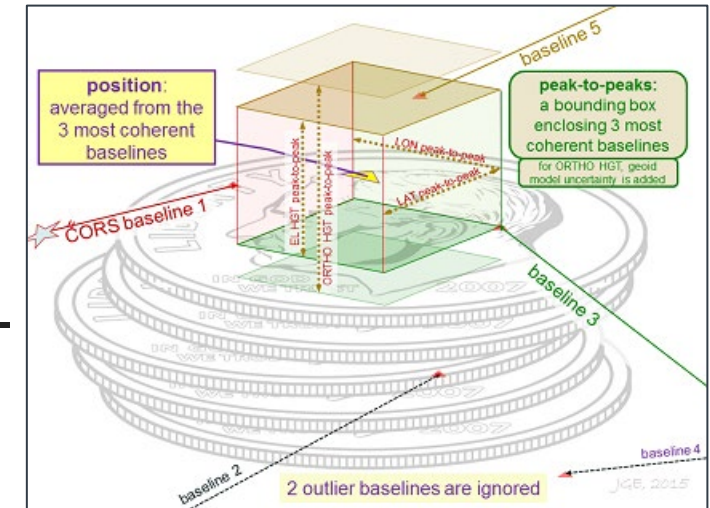
 Orthometric height (H): 0.747 (m)
 Std. Dev. H: 0.0121 (m)
 Geoid height (N): -36.118 (m)
 Std. Dev. N: 0.0116 (m)
 Geoid version: GEOID2022v1.0.0

ESTIMATED COORDINATES: SPCS2022 (EPOCH: 2020.0000)

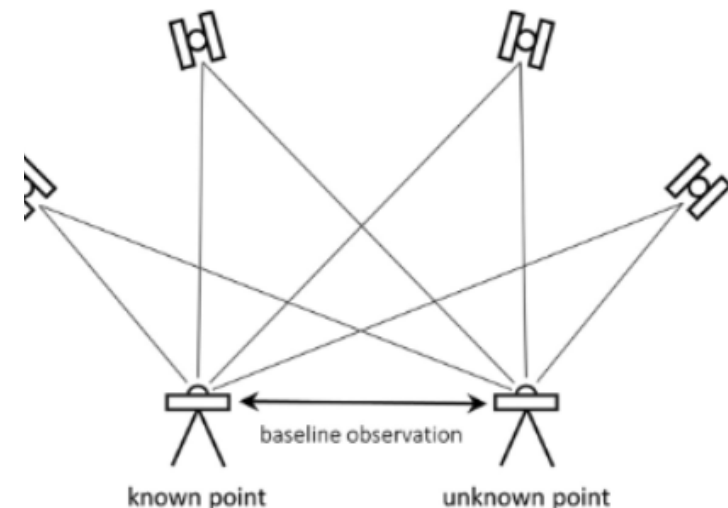
 SPC Zone: MD-240001
 Northing (N): 184,111.460 (m), 604,040.24 (ft)
 Easting (E): 876,973.145 (m), 2,877,208.57 (ft)
 Convergence: 0.55238056 (degree)
 Point Scale: 1.000004724
 Combined Factor: 1.000010274
 Linear distortion (ppm): 10.274

OPUS-S 5.2: Summary of Upgrades

- Multi-GNSS baseline processing
 - Simultaneous baseline processing
 - Error estimates are standard deviations, not peak-to-peaks
 - Full RINEX3 support ← recommended
- Occupations, $T \geq 1$ hour
- Enhanced base station selection
- Solutions in Modernized NSRS
 - SPCS2022 coordinates in meters and feet
- New JSON report
- Solutions can continue to be shared



Peak-to-Peak Errors Explained



Jamieson and Gillins (2018)

Development Phases for OPUS

OPUS-S 5.2

- **Purpose:** Update OPUS-S to give access to Modernized NSRS
- **New Features:**
 - Multi-GNSS
 - Solutions in Modernized NSRS
 - OPUS-Share supported
 - $T \geq 1$ hour
- **Status:** Development Completed, Internal testing

OPUS 6.0

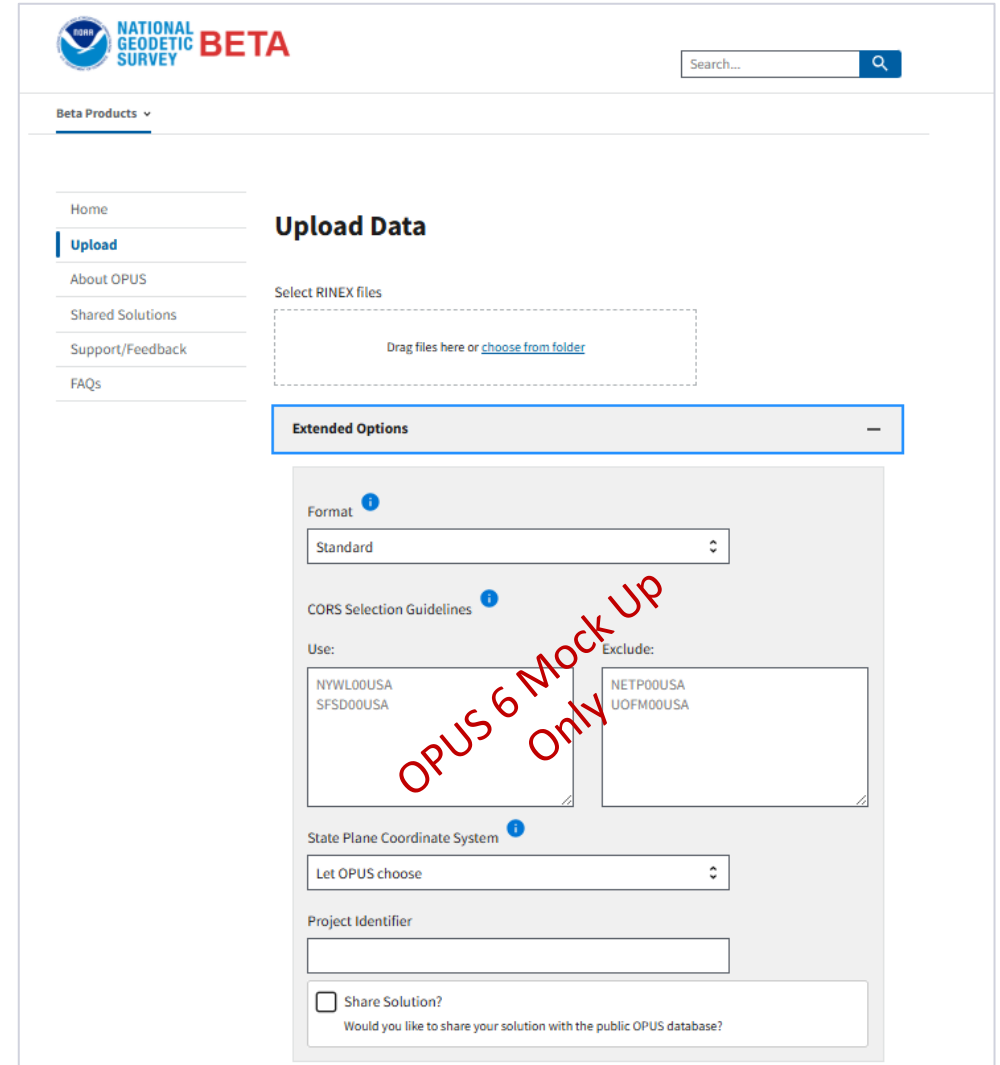
- **Purpose:** Re-write of OPUS in modern programming language
- **New Features:**
 - Cloud-native application
 - Single/multiple occupations on a mark
 - Preprocessing
 - Enhanced sharing functionality
- **Status:** Project plan & requirements completed; development underway

OPUS 6.1

- **Purpose:** Replacement for OPUS-Projects functionality
- **New Features:**
 - Multiple occupations on network of marks
 - Session processing customization
 - Network adjustments
- **Status:** Requirements gathering

OPUS 6.0: Core Development Requirements

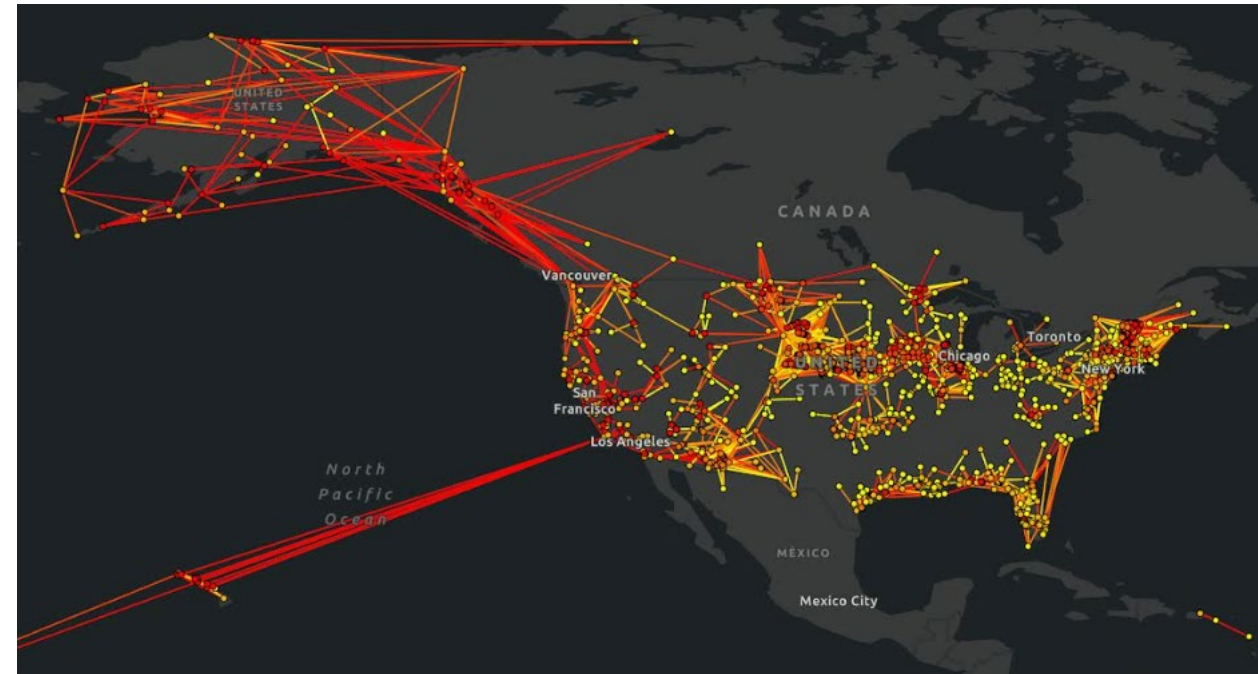
- Cloud-native architecture
 - scalability & reduced O&M
- Selects base stations according to recommendations
- Multi-GNSS (M-PAGES)
- Single occupation on a single mark
 - Pre-processes data file to list input antenna height & type, GNSS, file duration
- Multiple occupations on a single mark
 - Uploads ~10 data files on a mark
 - Selects common base stations for processing of baselines
 - Auto checks precision of the repeat measurements
 - Propagates measurements to reference epoch (2020.00) using IFDM2022
 - Performs least squares adjustment using LASER
 - Emails adjusted coordinates in a multi-occupation solution report



The screenshot shows the OPUS 6.0 Beta web interface. At the top is the NOAA National Geodetic Survey logo and a search bar. Below the logo is a navigation menu with links: Home, Upload, About OPUS, Shared Solutions, Support/Feedback, and FAQs. The main content area is titled "Upload Data" and includes a section for "Select RINEX files" with a dashed box for dragging files or a link to "choose from folder". Below this is an "Extended Options" section with a dropdown menu. The options include: Format (Standard), CORS Selection Guidelines (with a link to "CORS Selection Guidelines"), Use (NYWL00USA, SFSD00USA), Exclude (NETP00USA, UOFM00USA), State Plane Coordinate System (Let OPUS choose), Project Identifier, and a checkbox for "Share Solution?" with the text "Would you like to share your solution with the public OPUS database?". A large red diagonal watermark reading "OPUS 6 Mock Up Only" is overlaid on the interface.

OPUS 6.0: Enhanced “Sharing” Functionality

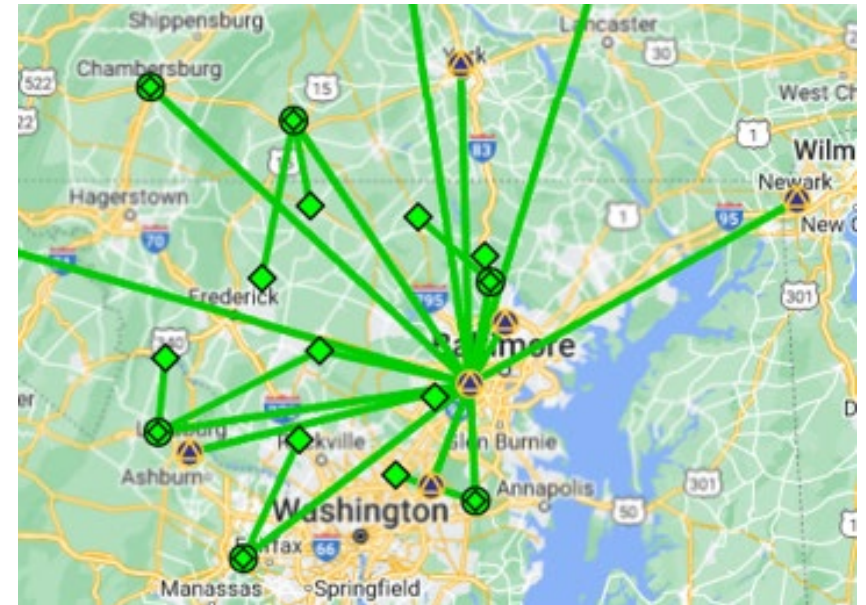
- Option to submit GNSS data to NGS
- Requirements:
 - $n \geq 2$ independent occupations
 - $T \geq 4$ hours
 - Meets data quality thresholds
 - recovery notes for mark
 - Close-up & profile photos
- NGS will review and add these data to the next REC (National) Adjustment for producing geodetic control coordinates



Shared OPUS Solutions, 5/2025 – 1/2026

OPUS 6.1

- Multiple occupations on a network of marks
 - Campaign-style surveys
- Session GNSS processing customization
- Network least-squares adjustments with LASER
- Submit projects to NGS for review & inclusion in next REC adjustment
- User training & outreach
- Built for future expansion to support other geodetic measurements (classical, leveling)



Data Delivery Service

Beta Products ▾

Switch Mark: DE8888 CHA 05A Print to PDF Export CSV Export JSON Export Text

Geodetic Mark Pages

DE8888 CHA 05A

NSRS Coordinates Hide

DE8888: CHA 05A

Epoch: 2020.0 Coordinates: Geographic Reference Frame: NATRF2022 Vertical Datum: NAPGD2022

Reference Epoch Coordinates

Reference Epoch Coordinates are suitable for geodetic control.

	Value	Std dev.	Units	Source	Date
Latitude (φ)	40.0255226202	± X.XXXXXXXX	decimal degrees	XXXXX	XX/XX/XXXX
Longitude (λ)	-88.3688038976	± X.XXXXXXXX	decimal degrees	XXXXX	XX/XX/XXXX
Latitude (φ)	N40° 01' 31.88143"	± X.XXXXXXXX	DMS	XXXXX	XX/XX/XXXX
Longitude (λ)	W088° 22' 07.69403"	± X.XXXXXXXX	DMS	XXXXX	XX/XX/XXXX
Ellipsoid Height (h)	178.837	± X.XXX	meters	XXXXX	XX/XX/XXXX
Orthometric Height (H)	Not Available	Not Available	meters	XXXXX	XX/XX/XXXX
Dynamic Height (H _d)	X.XXXX	± X.XXXX	meters	XXXXX	XX/XX/XXXX

Geopotential Coordinates

	Value	Std dev.	Units	Source	Date

Mark Information Hide

DE8888: CHA 05A

Primary Info Location Info

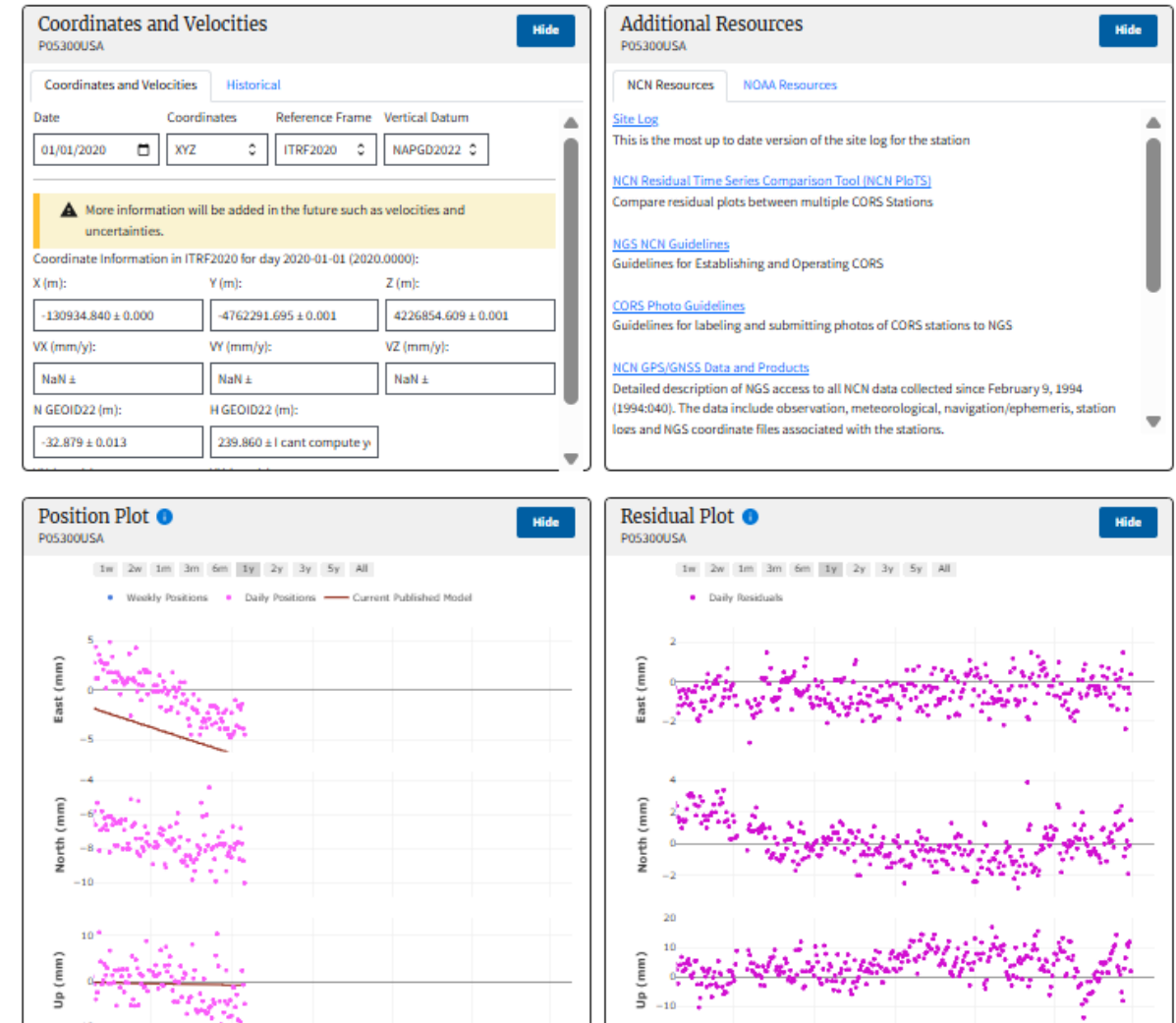
Designation	CHA 05A
Stamping	CHA-05A
Stability	B-Monument will probably hold position well
Setting	59-Stainless steel rod in sleeve (10FT+ or 3.048M+)
Monument Type	I-Metal rod
Rod Depth	3.7 meters
Sleeve Depth	1.1 meters
Magnetic Code	N-No magnetic material
GNSS Useable	Y
Setting Date	2000-09-29
Condition	G-Recovered in good condition
Last Recovered Date	2021-04-27

Photos Hide

DE8888: CHA 05A

No photos

Geodetic Mark Page

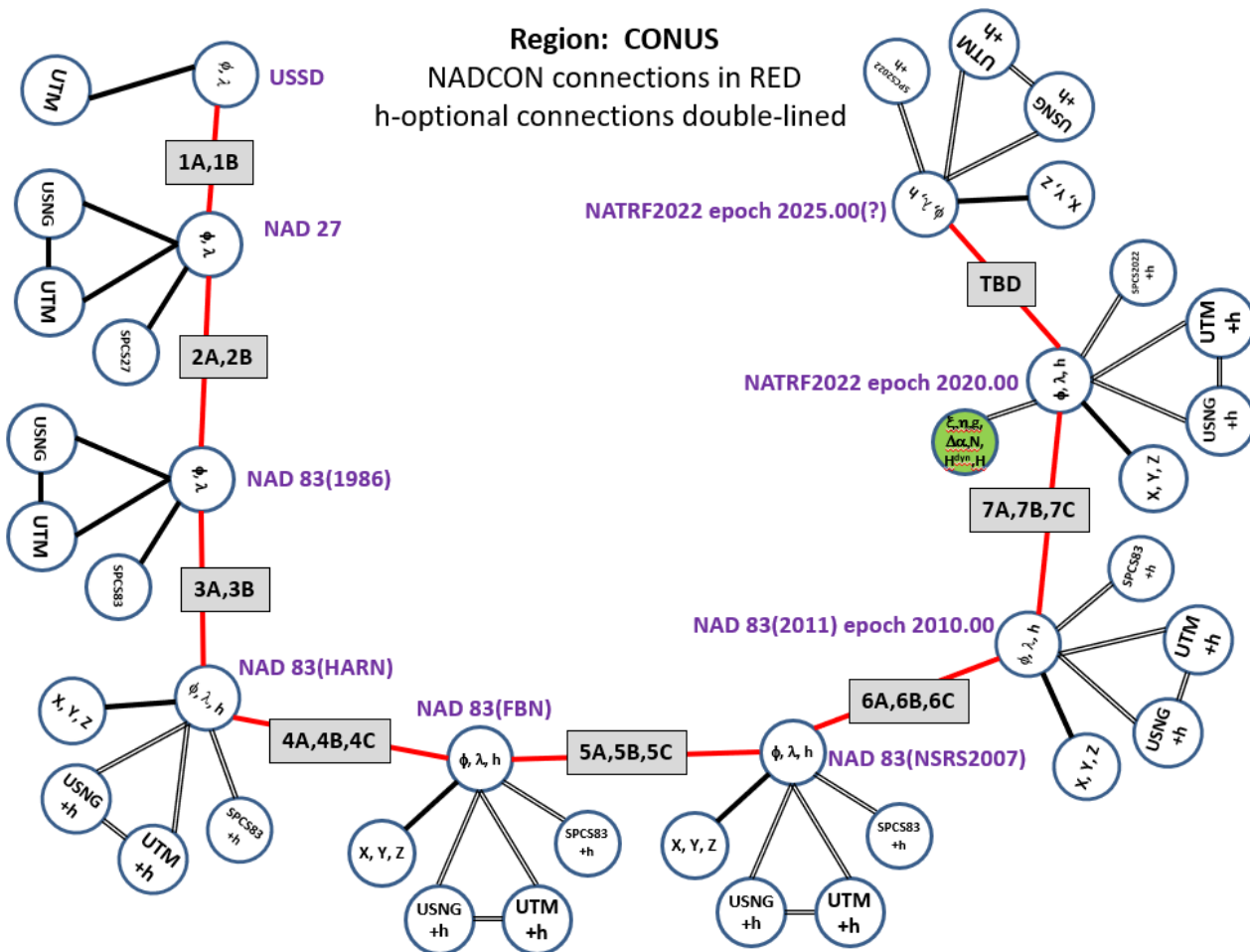


Geodetic Station Page

NCAT

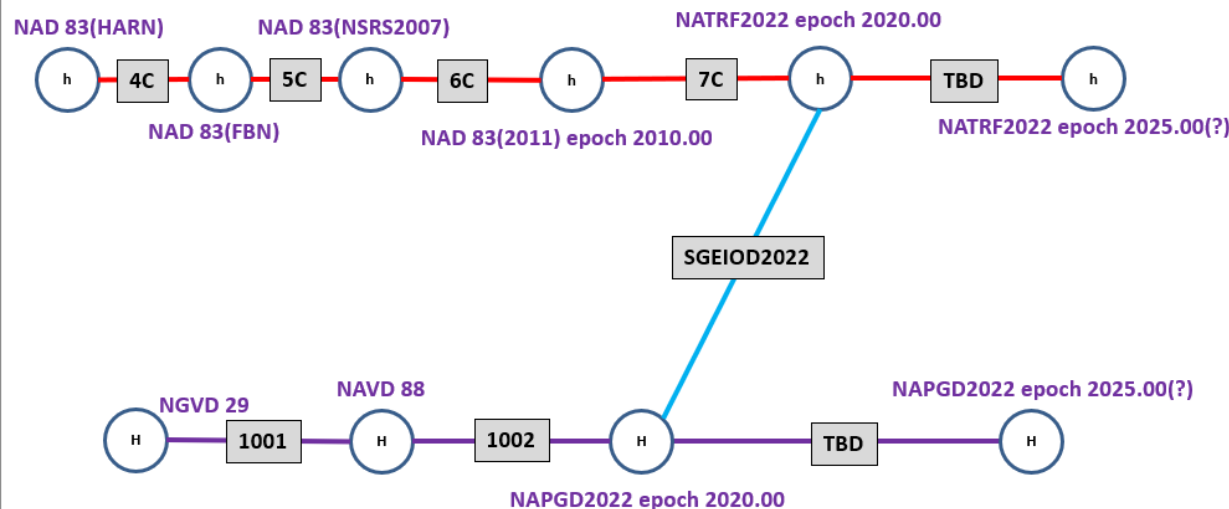
Latitude, Longitude

Region: CONUS
NADCON connections in RED
h-optional connections double-lined



Ellipsoid height, Orthometric Height

Region: CONUS
NADCON connections in RED
VERTCON connections in PURPLE
SGEIOD2022 connection in BLUE



Summary

- The modernized NSRS will be more accurate, accessible, efficient, and compatible with GNSS
- New terrestrial reference frames built upon ITRF2020
- NGS has 3 development phases for OPUS underway to support the Modernized NSRS
 - **OPUS-S 5.2:** Multi-GNSS with similar functionality as current OPUS-S
 - **OPUS 6.0:** Rewrite of code base in cloud-native application; single/multiple occupations on a single mark; enhanced sharing functionality
 - **OPUS 6.1:** Extension of OPUS 6.0 for multiple marks in a network; replace OPUS-Projects functionality